

A substrate recursion principle for biological information, with empirical anchoring through a templating-mode taxonomy

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(Dated: June 26, 2026)

Biological inheritance is a class of catalytic templating reactions: a daughter molecule, made by a catalyst acting on a parent template, is itself the template for the next round. We give the physicochemical conditions under which such a reaction transmits unbounded heritable distinguishability. Four conditions on the template–catalyst pair, for intrinsic non-teleological kernels, separate two necessity results. Linear scaling of substrate Shannon capacity with reaction extent requires positive per-site entropy, an unbounded count of independently variable positions, and catalytic closure under iteration, but not drift; a perfect copier transmits an exponentially large ensemble while producing no novelty. Diversification beyond the deterministic closure of a finite repertoire additionally requires stochastic drift in the recognised alphabet. A corollary organises six attested templating reactions into a taxonomy, and a finite-population proposition predicts bounded-heredity plateaus that simulation confirms. We test the framework on the bacterial defence system Drt3b, which makes alternating poly(AC) DNA without a nucleic acid template. It is a cyclic two-state channel with a one-bit ceiling; the framework predicts 10% dG misincorporation for its Glu26-to-Gln mutant against an observed 10.16%, and recovers a 15.7-fold dG elevation across six natural Glu26-to-Asp clades among 1,232 homologs. Two active-site substitutions provide single-experiment tests.

POPULAR SUMMARY

A bacterial defense protein called Drt3b, recently characterized in *E. coli*, synthesizes DNA with a strict alternating ACAC pattern without copying any template. Two conserved active-site residues, Glu26 and Arg253, are modeled as enforcing an alternating two-state catalytic cycle that selects which nucleotide enters at each step. This is sequence without a sequence template, and it does not fit the textbook picture of inheritance.

We treat inheritance as a class of catalytic templating reactions and ask which chemistries can support open-ended evolution. Two distinct requirements emerge. *Capacity*, the ability to transmit exponentially many distinct heritable configurations, requires three conditions on the template–catalyst pair: more than one monomer state recognized at each position, a position count that grows unboundedly with the reaction extent, and applicability of the catalysis to its own product. *Generation* of novel heritable configurations beyond what is already present requires a fourth condition: stochastic drift of the catalysis in its recognition alphabet. Capacity and generation are separate requirements. Faithful copying alone gives capacity without generation; the drift condition is what adds generation. Drt3b fails the second capacity condition, because its cycle has two states regardless of product length. The framework classifies six attested biological templating reactions as instances or partial instances of the same chemical specification, and it identifies two single-residue substitutions at Drt3b’s active site whose measured effects would test its predictions.

I. INTRODUCTION

A. The phenomenon and the puzzle

Biology runs on a small number of chemically distinct templating reactions for transmitting information across generations. Each is a catalyzed process in which a daughter molecule, generated from a parent template by the action of a catalyst, is itself the parent template for the next round of the same catalysis. DNA replication and ribosomal translation are the textbook cases: processive, ATP/GTP-coupled, polymerase- and ribosome-catalyzed monomer-by-monomer synthesis on a directing template. Prion propagation and amyloid templating add a second class, a conformational attractor propagated across protein molecules through nucleation-and-elongation chemistry [1]. Modular biosynthetic assemblies (non-ribosomal peptide synthetases, polyketide synthases) compose a third: a spatial sequence of catalytic modules executes a fixed sequence of condensation reactions, with the product specified by the active-site chemistry of each module [2]. The eukaryotic ciliate cortex constitutes a fourth: cortical patterns of basal-body orientation are inherited at division through the action of the existing cortical lattice on newly assembled units, and grafted inverted patches propagate as inverted for hundreds of cell generations without genomic change [3, 4]. The recent characterization of bacterial defense-related templating systems, the DRT family [5–7], adds at least one more. Drt3b synthesizes alternating poly(AC) DNA without using a nucleic acid template, generating its product through a cyclic active-site catalytic mechanism rather than through Watson-Crick complementarity to an existing strand [5].

These reactions pose a physicochemical question, not

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a purely biological one: what conditions on a template–catalyst pair are necessary for it to support unbounded heritable molecular distinguishability under repeated catalytic action? The mechanisms have unrelated active-site chemistries, but their dynamical signatures separate cleanly. DNA replication transmits a configurational ensemble whose Shannon capacity grows linearly with sequence length. Prion propagation transmits a finite conformational free-energy minimum and accumulates nothing comparable. The field has no quantitative chemical-physics account of which reactions sit on which side of this line, or on what physicochemical basis the line is drawn. The question is surprisingly absent from the current literature, despite half a century of attempts to address adjacent ones.

B. Prior work and the gap

The attempts have approached the substrate-level question from three directions. The information-theoretic line [8–10] treats copying-with-variation as a process whose stability depends on per-site error rate relative to genome length, generating the quasispecies framework and the error threshold $\mu^* \approx \log \sigma / L$. This work tells us how a substrate-borne replication process can fail dynamically, but not what makes a physical system a substrate at all. Szathmáry [10] introduced the distinction between limited heredity (a substrate with K states transmits a $\log_2 K$ -bit ceiling) and unlimited heredity (a sequence-based substrate transmits $L \log_2 |\mathcal{A}|$ bits, scaling with length). The distinction is correct in spirit, but it is asserted as a typological observation rather than formalized as a structural condition on substrates.

A second line [11, 12] specifies abstract requirements on a replicator (faithful self-copying, heritable variation in fitness) without committing to any chemistry that realizes them. It is chemistry-blind: it does not say which physical realizations of the abstract requirements actually occur, or through what catalytic mechanism. The substrate-level line has attempted to specify the physical substrate directly. Schrödinger’s aperiodic crystal [13] was prescient but not formalized. The GARD-class compositional-inheritance models [14, 15] proposed that a compositional set of catalytic molecules transmits information across generations through preferential reproduction of similar compositions; Vasas *et al.* [16] showed, using GARD’s own machinery, that compositional inheritance fails the fidelity requirements for open-ended evolution. Pross [17] supplied a thermodynamic condition (dynamic kinetic stability) on replicators sustained far from equilibrium.

What is missing from all three lines is a single statement, at the chemical-physics level of the template–catalyst pair, of what conditions are necessary for a templating reaction to transmit unbounded heritable distinguishability. The information-theoretic work treats pop-

ulation dynamics on a reaction taken as given, not the reaction itself. The replicator-criteria work stays abstract about the chemistry. The substrate-level attempts have so far either gestured at the answer, as in Schrödinger, or been refuted on their own terms, as with GARD. The Drt3b finding sharpens the gap. Drt3b is a biological templating reaction that is not a sequence-template polymerase, and a framework that classifies it has to operate at the level of the catalytic mechanism itself.

C. Here we show

We give such a framework. Four structural conditions on the template–catalyst pair, together with a fifth scope-defining condition, separate two necessity results: a capacity theorem and a generation theorem. Capacity is storage architecture under the catalytic kernel; generation is stochastic branching to neighboring configurations under the same kernel. Both are necessary for biology’s open-ended evolution, and they have different chemical requirements. A bounded-heredity corollary organizes six attested templating reactions into a taxonomy keyed to which condition each reaction fails, and a finite-population proposition characterizes the population dynamics each class hosts.

Section II states the framework: the formal definitions, the five conditions, the two theorems with proofs, the bounded-heredity corollary and its mode taxonomy, and the finite-population proposition. Section III reports the work that tests the framework: a descriptor-relative diagnostic apparatus, mode-specific scaling laws, quantitative anchoring on the Drt3 reverse-transcriptase system, a family-level analysis across 1,232 Drt3b homologs, site-directed mutagenesis predictions, and a finite-population simulation, followed by the framework’s relationship to prior traditions, its limitations, and the orthogonal preconditions of life that lie outside its scope. Section IV states the conclusions. Materials and Methods give the formal setup, the simulation parameters, and the test protocols.

A note on vocabulary. We use *drift* rather than *error* or *variation* for the stochastic part of the catalytic kernel: the kernel produces what its physics produces, and there is no externally imposed target configuration the catalysis is trying to reach. We retain *error* for established field usage (Eigen’s threshold, and the per-position misincorporation rate ε defined against a Watson–Crick canonical product), where the technical meaning is fixed.

II. DEFINITIONS, CONDITIONS, AND THEOREMS

We state the framework in three parts: definitions of heredity capacity and cumulative generation, the substrate recursion conditions, and two necessity theorems that separate capacity from generation. Each formal

statement is preceded by a plain-language framing of what it claims and followed by a biological example.

a. Heredity capacity and cumulative generation. Before specifying conditions on the substrate, we specify what the substrate is required to support. “Open-ended information generation” has been used loosely, and the looseness hides a quantifier ambiguity that determines which structural conditions are necessary. We separate two formal notions.

The first is *substrate heredity capacity*: how much information the substrate can transmit per recursion event, treated as an encoder property. The capacity is the supremum of mutual information over input ensembles supported on the recursively closed set $\mathcal{R}_L := \{X \in \mathcal{X}_L : \mathcal{O}^t(X) \in \mathcal{X}_L \text{ for all } t \geq 0, \text{ possibly via } \iota\}$:

$$C_S(L) = \sup_{\Pi_X \text{ on } \mathcal{R}_L} I(\varphi_X(X); \varphi_X(X')), \quad X' \sim Q_L(\cdot | X). \quad (1)$$

Restricting the supremum to $\mathcal{R}_L$ makes R3 (catalytic closure under iteration) mandatory in the definition. A substrate failing R3, such as translation, where the protein product is not ribosome-readable as a next-round template, has $\mathcal{R}_L = \emptyset$ or trivially small, so $C_S(L)$ is zero or bounded by a constant. This distinguishes recursive heredity capacity from one-step channel capacity, which can be $\Omega(L)$ for non-recursive transfers (DNA $\rightarrow$ protein carries $\Omega(L)$ one-step bits but transmits them once and exits the substrate class). The substrate has *unlimited heredity capacity* if $C_S(L) = \Omega(L)$; equivalently, in the discrete noiseless limit, if the set of heritably distinguishable and recursively readable configurations $\mathcal{R}_L$ satisfies $|\mathcal{R}_L| \geq 2^{cL}$ for some $c > 0$.

Definition 1 (Heredity capacity). A substrate-operation pair $(\mathcal{X}, \mathcal{O})$ has unlimited heredity capacity if $C_S(L) = \Omega(L)$ as $L \rightarrow \infty$. Equivalently, in the discrete noiseless limit, $|\mathcal{R}_L| \geq 2^{cL}$ for some $c > 0$.

Remark 1. The two formulations are coordinated through entropy: linear scaling of mutual information capacity ($C_S(L) = \Omega(L)$ bits) is equivalent to exponential scaling of the recursively readable state count ($|\mathcal{R}_L| = \Omega(2^{cL})$). The first is the information-theoretic statement; the second is the combinatorial statement. They are the same fact at two levels of description.

The second is *cumulative generative recursion*: the substrate’s ability, starting from a finite initial repertoire $S_0 \subsetneq \mathcal{X}_L$ small compared to its reachable set, to produce heritable configurations *beyond the zero-drift closure* of S_0 under $\mathcal{O}$, through branching local alternatives in the same recognized alphabet. The qualifier “beyond the zero-drift closure” is essential. A deterministic operation can leave S_0 (a counter chain $000 \rightarrow 001 \rightarrow 010 \rightarrow \dots$ generates configurations outside any singleton S_0), but the descendant set is then a single predetermined orbit. Cumulative generation in the chemical-kinetics sense requires that the future is not pre-specified: the kernel produces heritable branching alternatives in Σ at each step, so the descendant set is genuinely larger than the orbit

closure under the catalysis’ canonical zero-drift output.

Formally, let $g_L(X) := X_{\text{ref}}(X)$ be the canonical zero-drift map of $\mathcal{O}$ (the kernel’s reference output: the Watson–Crick complement for a DNA polymerase, the exact alternating product for Drt3b). Let $\text{cl}_0(S_0) := \bigcup_{t \geq 0} g_L^t(S_0)$ be the orbit closure of S_0 under zero-drift iteration. Then $(\mathcal{X}, \mathcal{O})$ supports cumulative generation if the recursively readable descendant set reachable from S_0 with positive probability under Q_L strictly contains $\text{cl}_0(S_0)$ for all proper $S_0 \subsetneq \mathcal{X}_L$ with $|S_0| \ll |\mathcal{R}_L|$, with the additional configurations differing from elements of $\text{cl}_0(S_0)$ in local recognized-state positions (in the alphabet Σ). Using the canonical zero-drift skeleton avoids the pitfall of defining closure as a union over all deterministic restrictions of Q_L : for stochastic Q_L , such a union absorbs every positive-probability branch and trivializes the strict-containment condition. The zero-drift formulation picks one canonical deterministic skeleton, so R4-failure (zero drift, $Q_L = g_L$) gives equality rather than impossibility.

Definition 2 (Cumulative generative recursion). A substrate-operation pair $(\mathcal{X}, \mathcal{O})$ supports cumulative generative recursion if, for arbitrarily large L and any proper finite initial repertoire $S_0 \subsetneq \mathcal{X}_L$ with $|S_0| \ll |\mathcal{R}_L|$, repeated application of Q_L generates a recursively readable descendant set whose heritable descriptor support strictly contains the zero-drift closure $\text{cl}_0(S_0)$, with the additional configurations arising through branching local alternatives in the same recognized alphabet Σ as in (R1).

DNA illustrates both. The 4^L -configuration space gives unlimited heredity capacity; the polymerase’s drift in the nucleotide alphabet gives cumulative generation. The two are conceptually independent. A hypothetical perfect copier of DNA satisfies Definition 1 (any of 4^L templates is faithfully preserved across iterations) but not Definition 2 (the descendant set from a finite S_0 equals $\text{cl}_0(S_0)$ itself, with no branching alternatives in Σ). A purely deterministic-counter substrate likewise fails Definition 2: the descendant set is bounded by the zero-drift orbit. Drt3b fails Definition 1 at the substrate level: at any product length, its two-phase cyclic dynamics distinguish only $\log_2 2 = 1$ bit of phase, so $C_S(L) = O(1)$.

A note on what the kernel encodes. $Q_L(X' | X)$ is the distribution over outcomes that the active-site free-energy landscape, the available reactants, and the chemical driving (NTP hydrolysis or an analogous energy source) produce when the catalysis acts on X . The kernel does not encode a target product; every template in iterated Q_L -application is the drifted product of prior runs of the same reaction, and no configuration is canonical at the kernel level.

b. The substrate recursion conditions. The substrate-operation pair $(\mathcal{X}, \mathcal{O})$ is the formal object the principle constrains. $\mathcal{X}_L$ is the space of admissible configurations of size L ; $\varphi_X(X) \in \Sigma^L$ is the descriptor of X over the recognition alphabet Σ ; $\mathcal{O}$ is the operation

with kernel $Q_L(X' | X)$ producing $X' = \mathcal{O}(X)$, possibly composed with a reversible involution $\iota : \mathcal{X}_L \rightarrow \mathcal{X}_L$. The involution covers the canonical case where the immediate product of $\mathcal{O}$ is the complement of the input (Watson-Crick complementarity produces $\bar{X}$, and the next round of replication on $\bar{X}$ recovers X), allowing the recursion to close even when each individual $\mathcal{O}$ application produces a complement rather than a copy. Four conditions are structural and are what the necessity theorems require; the fifth is scope-defining.

Definition 3 (Substrate recursion conditions). The substrate-operation pair $(\mathcal{X}, \mathcal{O})$ satisfies the substrate recursion conditions if:

(R1) Positive entropy per recognized site.: At

each addressable position i , the recognition step in $\mathcal{O}$ distinguishes at least two alternative monomer states drawn from a recognition alphabet Σ with $|\Sigma| \geq 2$. The per-site entropy under the maximum-entropy distribution, $h_{\text{site}} = \log_2 |\Sigma|$, is positive.

(R2) Unbounded independently variable position count.:

The substrate has non-recurrent sites whose count $N(L)$ grows unboundedly: $\liminf_{L \rightarrow \infty} N(L) = \infty$. “Independently variable” is meant in the first-order, descriptor-relative sense: increasing L increases the count of heritable recognized degrees of freedom under the operation-relevant descriptor, and the substrate is not constrained to a finite recurrent phase repertoire. The condition does not require recognized states at distinct positions to be statistically independent in any strong joint-distribution sense; DNA polymerases exhibit context-dependent error rates and base-composition biases, and DNA still satisfies R2 because the count of independently variable positions grows with sequence length. Composition vectors, homogeneous concentrations, and bulk mixtures fail R2 because they lack addressable position. Cyclic systems with finite recurrent phase repertoires fail R2 because $N(L)$ is bounded at the cycle length: product length can grow without bound while the count of heritable degrees of freedom stays at the cycle complexity N . This sharpening protects the Drt3b argument from a naive “product length grows” reading.

(R3) Same-operation recursion (with allowed involution).:

The output $X' = \mathcal{O}(X)$ is structurally an element of $\mathcal{X}_L$ for which $\mathcal{O}(X')$ is well-defined, possibly via a reversible involution ι such that $\mathcal{O}(\iota(X')) \in \mathcal{X}_L$. The recursion takes the form

$$X \xrightarrow{\mathcal{O}} X' \xrightarrow{\mathcal{O}} X'' \xrightarrow{\mathcal{O}} \dots \quad \text{or} \quad X \xrightarrow{\mathcal{O}} \iota(X') \xrightarrow{\mathcal{O}} X'' \xrightarrow{\mathcal{O}} \dots$$

The condition excludes operations whose products are not readable by the same operation.

(R4) Heritable local drift in the recognized alphabet.:

The kernel $Q_L(X' | X)$ has nonzero probability of producing configurations $X' \neq X_{\text{ref}}(X)$ for any reference configuration, with the drift lying in the same alphabet Σ as the recognition step in (R1). Substrate content (the descriptor $\varphi_X(X)$) and substrate templating function (the role of X in $\mathcal{O}(X) \rightarrow X'$) are the same physical entity: drift in content is drift in the next iteration’s template.

(R5) Scope: non-teleological intrinsic operation.:

The kernel Q_L depends only on the physicochemistry of $(\mathcal{X}, \mathcal{O})$, not on an externally specified target configuration, loss function, or convergence criterion. (R5) is scope-defining rather than structural: it demarcates the domain in which (R1)–(R4) apply. Differential persistence of configurations may occur downstream over a finite population, but selection acts on products, not on the operation itself. Systems whose operation is externally optimized (gradient descent on a loss function, designed catalytic cycles with specified targets) are outside the principle’s scope.

The conditions in biological terms:

- **(R1)** requires the recognition step to distinguish at least two states at each site. DNA satisfies this: at each position the polymerase distinguishes A, T, G, and C, giving $h_{\text{site}} = 2$ bits. A polymerase that incorporates only T has $h_{\text{site}} = 0$ and fails (R1).
- **(R2)** requires the count of independently variable sites to grow without bound. DNA satisfies this; Drt3b’s cycle has only two phases regardless of product length, so its site count is bounded at 2 and (R2) fails.
- **(R3)** requires the operation to apply to its own output. DNA replication satisfies this through the involution $X \rightarrow \bar{X}$. Ribosomal translation fails (R3): the protein product is not a substrate the ribosome can re-read, and recursion is recovered only through Mode 1 acting on the gene encoding the ribosomal machinery.
- **(R4)** requires drift in the alphabet the operation recognizes. DNA polymerases produce drift in the nucleotide alphabet (a $\sim 10^{-9}$ per-position rate against the Watson-Crick canonical product), and the drifted product is itself recognized at the next iteration. A perfect copier fails (R4); so does a system that drifts in an alphabet the polymerase does not read.
- **(R5)** requires intrinsic physicochemistry. DNA replication satisfies it; a neural network trained by gradient descent toward a target distribution does not, and is therefore outside the principle’s scope. (R5) does not say the externally optimized system fails biology; it says the principle’s structural analysis does not apply to it.

c. *Two necessity claims.* The two definitions yield two distinct necessity claims. The first concerns heredity capacity, the substrate’s encoder properties as an information channel. The second concerns cumulative generation, its ability to innovate from a finite seed. Both are necessary for biology’s open-ended evolution, but they have different structural requirements, and the empirical phenomena are clearer when the two are kept separate.

Theorem 1 (Capacity: necessity of R1, R2, R3 for unlimited heredity capacity). *Within the scope defined by (R5), if a substrate-operation pair $(\mathcal{X}, \mathcal{O})$ has unlimited heredity capacity in the sense of Definition 1, then it satisfies (R1), (R2), and (R3).*

Theorem 2 (Generation: necessity of R4 for cumulative generative recursion). *Within the scope defined by (R5), if a substrate-operation pair $(\mathcal{X}, \mathcal{O})$ supports cumulative generative recursion in the sense of Definition 2 (its descendant set from finite S_0 strictly contains $\text{cl}_0(S_0)$ through branching local alternatives in Σ), then it satisfies (R4) in addition to (R1), (R2), and (R3).*

d. *Proof of Theorem 1 (sketch).* The argument is an upper-bound contrapositive. For any substrate-operation pair,

$$C_S(L) \leq N(L) \cdot \log_2 |\Sigma|, \quad (2)$$

because $C_S(L) = \sup_{\Pi_X} I(\varphi_X(X); \varphi_X(X'))$ is bounded by the entropy of the descriptor, which has at most $|\Sigma|^{N(L)}$ possible values. A matching lower bound requires a realizability condition: increasing L must actually create additional heritable recognized degrees of freedom under the operation-relevant descriptor. This is the operational content of R2, and it is what distinguishes a substrate with genuinely extensible recognized positions from one (a finite recurrent phase cycle) where product length grows but heritable degrees of freedom do not. Failure of (R1) gives $|\Sigma| = 1$, so $\log_2 |\Sigma| = 0$ and $C_S(L) = 0$. Failure of (R2) gives $N(L) \leq N_{\max}$ bounded, so $C_S(L) \leq N_{\max} \cdot \log_2 |\Sigma|$, constant in L and therefore not $\Omega(L)$. Failure of (R3) means $\mathcal{O}^2$ is not defined within $\mathcal{X}_L$: the iteration terminates at the first application or transitions to a different substrate class on the second, so capacity inherited via a different upstream substrate counts as parasitic recursion rather than $(\mathcal{X}, \mathcal{O})$ ’s own capacity. The contrapositive in each case gives the theorem. R4 is not invoked and is not necessary for capacity: a substrate satisfying R1, R2, R3 with $Q_L(X' | X) = \delta(X' - X)$ (a perfect copier) has $C_S(L) = N(L) \cdot \log_2 |\Sigma|$ and unlimited heredity capacity. This is why DNA without mutation would still be a high-capacity inheritance carrier: it could store and transmit its configurations even though it could not innovate.

e. *Proof of Theorem 2 (sketch).* R1, R2, and R3 are inherited from the requirement that the pair be coherent and recursively closed. Beyond these, suppose R4 fails. There are two sub-cases. (a) The kernel is deterministic with $Q_L(X' | X) = \delta(X' - g_L(X))$, where $g_L(X) = X_{\text{ref}}(X)$ is the canonical zero-drift map. Then

for any S_0 , the descendant set after T iterations is $\bigcup_{X_0 \in S_0} \{X_0, g_L(X_0), \dots, g_L^T(X_0)\}$, which equals $\text{cl}_0(S_0)$. The deterministic operation may leave S_0 , but the descendant set is contained in the zero-drift orbit closure, with no branching alternatives, so strict containment fails. (b) The kernel produces drift in an alphabet Σ' disjoint from the recognition alphabet Σ : the drifted configurations are not re-recognized at the next iteration, so they do not propagate as recursively readable descendants. Definition 2 requires the new configurations to arise through branching in Σ itself. In both sub-cases the substrate cannot produce a recursively readable descendant set strictly containing $\text{cl}_0(S_0)$ through branching in Σ , contradicting Definition 2.

The “beyond deterministic closure” qualifier is the crucial logical move. An earlier formulation of the generation theorem said only that R4 is necessary for the descendant set to leave S_0 , which is false: a deterministic counter chain leaves any singleton S_0 without satisfying R4. What R4 is actually necessary for is the descendant set being larger than the deterministic orbit closure, that is, the substrate producing genuine branching alternatives at each step rather than a single predetermined trajectory. The full proofs, including the argument that the conjunction’s scaling rate cannot be recovered from any subset of conditions, are in Supplementary Mathematical Appendix §5.

Theorem 1 explains why bounded-capacity substrates (Drt3b, prions) cannot support open-ended inheritance regardless of how much drift they produce: their structural ceiling caps the heritable repertoire at the wrong scaling. Theorem 2 explains why a high-capacity substrate without drift cannot innovate from a finite seed: with no branching in the recognized alphabet, the descendant set never exceeds the deterministic closure of that seed. Biology’s DNA-based heredity satisfies both: 4^L scaling gives capacity, and $\sim 10^{-9}$ per-position polymerase drift gives generation. The behaviors standardly described as natural selection, fitness, error thresholds, and evolutionary plateaus are downstream consequences, which we treat next as a corollary and a scoped proposition.

A. Bounded heredity and the substrate-mode taxonomy

A substrate failing one of (R1)–(R4) cannot reach the corresponding open-ended regime, but it is not thereby left with no heredity. What fails is the unboundedness of transmission (capacity, if R1, R2, or R3 fail) or the generative reach beyond a finite repertoire (if R4 fails). The heredity is bounded, and the bound is set by which condition fails.

Corollary 1 (Bounded heredity). *If $(\mathcal{X}, \mathcal{O})$ is in the scope of (R5) and satisfies a strict subset of (R1)–(R4), then the substrate may exhibit heredity, but the heredity is bounded in a manner determined by which conditions*

fail. Substrates failing (R1) have configuration-space ceilings set by the per-site entropy ($h_{\text{site}} = 0$ at the limit, $h_{\text{site}} \leq \log_2 K$ for a K -state attractor); substrates failing (R2) have ceilings set by bounded position count N_{max} (cycle length, module count); substrates failing (R3) inherit parasitically through an upstream substrate that satisfies (R3); substrates failing (R4) exhibit fixed configurations or non-recursive drift without information growth.

The corollary’s empirical instantiation is biology’s substrate-mode inventory. We document six modes, each a specific failure pattern of (R1)–(R4); capacity bounds and proofs are in Supplementary Mathematical Appendix §2–3.

Mode 1: complementarity-based 1D sequence templating. Substrate: a 1D linear sequence on a separate molecule. Mechanism: position-by-position chemical complementarity (Watson-Crick, β -sheet hydrogen bonding). Satisfies all four structural conditions within (R5) scope: $h_{\text{site}} = \log_2 |\mathcal{A}| > 0$, unbounded position count $N(L) = L$, daughter strand a reactant for the next replication via complementarity (R3 via involution $\iota : X \rightarrow \bar{X}$), and heritable local drift through the polymerase kernel. Capacity: $C_S(L) = L \cdot \log_2 |\mathcal{A}|$ in the noise-free limit. Examples: DNA replication, Drt3a, retrons, telomerase, DRT9, β -sheet-mediated peptide replication.

Mode 2: code-encoded cross-descriptor templating. Substrate: a 1D sequence in X plus separable lookup-table machinery in $\mathcal{O}$. Fails (R3) at the substrate level: protein products are not readable by the ribosome as templates for further translation, so the recursion is recovered parasitically through the encoding gene, which is itself a Mode 1 substrate. Information transfer under uniform codon input is bounded by $I_{\text{struct}}^{\text{POP}} \leq L \cdot H(\nu)$, where ν is the codon-count distribution over output symbols ($H(\nu) = 4.218$ bits/codon for the standard genetic code, including stop); the recursive capacity C_S itself is near zero because R3 fails. Example: ribosomal translation.

Mode 3: cyclic-structure-encoded templating. Substrate: a 3D protein structure with cyclic conformational dynamics having N distinct states. Fails (R2) in the sharpened sense: although product length can grow without bound, the count of independently variable recognized positions equals the cycle phase, bounded at N and recurrent ($N(L) = N$ regardless of product length). The macro-level capacity is consequently bounded at $\log_2 N$, and the output is periodic with period N . This distinguishes the framework’s R2 from a naive “product length grows” reading: Drt3b produces arbitrarily long products, but its heritable distinguishable information stays at $\log_2 N$ bits. Examples: Drt3b producing $(\text{AC})_n$ via $N = 2$ conformational states [5]. DRT7 is the boundary case at $N = 1$, where $\log_2 1 = 0$: there is no positional information capacity, only structural channel selectivity, so we classify DRT7 as a Mode 3 boundary rather than a positional-templating instance.

Mode 4: conformer state-templating. Substrate:

a discrete conformational state of a protein. Fails (R1) at the macro level (the recognition alphabet is a finite attractor repertoire of K conformers) and fails (R2) trivially (a single effective position, the conformer assignment, rather than an extensible series). Capacity: $C_S(L) \leq \log_2 K$, length-independent. Examples: prion propagation, amyloid self-conversion, DRT1 filamentous oligomer [6]. Prion strain repertoires confirm that some drift is transmitted across iterations, but the transmissible drift lies in a small attractor set rather than an extensible alphabet.

Mode 5: modular conveyor templating. Substrate: a spatial sequence of N structurally distinct modules within a single multi-modular assembly. Fails (R2) at the module count (position count is bounded by N , not extensible without adding modules) and (R3) at the substrate level (peptide products are not the assembly line; recursion is parasitic on Mode 1 acting on the gene cluster). Capacity: $C_S(L) = N \cdot \log_2 k$ for $L \geq N$, with output length bounded by N . Examples: non-ribosomal peptide synthetases, polyketide synthases.

Mode 6: 2D surface-position templating. Substrate: a spatial register on a 2D pattern, where the existing pattern biases the assembly of new units that do not themselves store the pattern. Mode 6 has extensible geometric positions (R2 holds: surface position count grows with area) and per-site entropy in the unit-type alphabet (R1 holds with multiple unit types). The failures are at (R3) and (R4): the 2D pattern does not autonomously self-copy with heritable local drift independent of genome-encoded parts. Pattern copying is not a same-operation recursion of the surface; it depends on Mode 1 acting on the genome to produce the constituent proteins, and pattern drift independent of the genome’s drift cannot accumulate as heritable variation in the recursive sense. The biological instance is the eukaryotic ciliate cortex [3, 4]: cortical patterns of ciliary basal-body orientation are inherited at division through templating action of the existing cortex on newly assembled units, and microsurgically grafted inverted patches propagate as inverted for hundreds of generations without genomic change. The framework retrodicts the cortex as a real but bounded inheritance system, parasitic on Mode 1 in the same structural sense as Mode 2.

The taxonomy is an *empirical inventory* of currently analyzed biological templating architectures, not a deductive exhaustion of all possible substrates satisfying or partially satisfying (R1)–(R4). Every characterized DRT system fits one of Modes 1, 3, 4, plus the framework’s out-of-scope boundary (Tables II–III); Modes 2, 5, 6 are anchored elsewhere (translation, NRPS/PKS, ciliate cortex). The inventory is provisional in the sense any empirical taxonomy is: new biology may force additions, as Drt3b forced Mode 3 to be made explicit. A more principled future taxonomy might derive modes from substrate dimensionality, recognition cardinality, and recursion topology as orthogonal axes within the principle.

B. Finite-population proposition

The structural conditions classify substrates; the population dynamics those substrates host follow as a scoped, finite-horizon claim. An earlier formulation stated a strong universal claim, that R3- or R4-failing mechanisms are structurally capped at mechanism-specific fitness ceilings under any population dynamics. That claim is too strong: a wholesale-redraw mechanism with full support over the configuration space can in principle sample the target by chance given infinite time, even though the probability is negligible in any finite regime. The issue is efficiency, not absolute reachability, so we state the result as a proposition scoped to the tested finite-population landscape model and framed as an adaptive-efficiency argument. The proposition is tested by simulation in Sec. III F.

Proposition 1 (Finite-horizon plateau under nonlocal or absent drift). *In a finite population of size K , evolving under selection sharpness β toward a target phenotype on the tested-class landscape (single-target Hamming, L -dimensional alphabet, static), over a horizon $T < \infty$:*

1. *Mechanisms whose substrate-operation pair fails (R4) at the individual level (exact-copy systems) cannot exceed the maximum fitness present in their founding lineage repertoire. They are bounded by initial-population properties, not by their own variation capacity.*
2. *Mechanisms with nonlocal (R4) (wholesale redraw at rate $r > 0$) sample novelty with full support but do not accumulate local improvements; under the tested update rules, the probability of reaching $F_{\max}$ within horizon T is dominated by random hitting time on the full configuration space and remains negligible for $L \gg 1$.*
3. *Mechanisms satisfying (R3) and (R4) with local drift at the individual level are not structurally barred from cumulative climbing toward $F_{\max}$. Whether climbing actually occurs depends on landscape topology, finite-population drift, mutation load, neutral-network connectivity, and other conditions outside the principle's scope.*

In words: exact-copy mechanisms cannot improve past the fittest variant already present at founding, because selection can only sort the standing repertoire and adds nothing to it. Wholesale-redraw mechanisms do introduce variation, but each redraw discards the lineage's accumulated state, so local improvements rarely compound within the horizon. Only individual-level local drift feeds variation into selection in a form that accumulates. This positive part is bounded: local-drift mechanisms are not structurally barred from climbing, though whether they climb depends on the landscape.

III. RESULTS AND DISCUSSION

A. A descriptor-relative apparatus distinguishes templating from biasing participants

The framework's empirical anchoring requires an instrument that classifies templating events by mode. We define a templating event as the tuple $(X, \mathcal{O}, S, \Delta G; \varphi_X, \varphi_Y) \rightarrow Y$, where X is the template, $\mathcal{O}$ the operator, S the substrate pool, ΔG the free-energy driving, and φ_X, φ_Y the descriptor maps. The diagnostic apparatus is a joint observable in three coordinates.

The first is *population mutual information*,

$$I_{\text{struct}}^{\text{pop}}(X; Y) = H(\varphi_Y(Y)) - \mathbb{E}_{X \sim \Pi} [H(\varphi_Y(Y) | X)], \quad (3)$$

the expectation taken over a stated template population $\mathcal{X}$ with distribution Π . It is bounded above by $H(\Pi)$ and is identically zero when $\mathcal{X}$ is degenerate, so it measures *realized* information transfer in a stated population, distinct from substrate capacity.

The second is *channel mutual information* against a stated comparison ensemble $\mathcal{C}$ of templating channels,

$$I_{\text{struct}}^{\text{chan}}(\mathcal{C}) = \sum_{c \in \mathcal{C}} P(C = c) D_{\text{KL}}(P(\varphi_Y(Y) | C = c) \| \bar{P}(\varphi_Y(Y))), \quad (4)$$

with mixture distribution $\bar{P}(\varphi_Y(Y)) = \sum_c P(C = c) P(\varphi_Y(Y) | C = c)$. Unlike $I_{\text{struct}}^{\text{pop}}$, $I_{\text{struct}}^{\text{chan}}$ does not vanish for fixed-template channels: a channel with a single fixed template still produces a distinguishable output distribution against alternative channels.

The third is *per-base mechanism fidelity*, resolved into two components when the channel has cyclic structure. *Phase fidelity* f_{phase} is the fidelity of cycle-state alternation; *monomer fidelity* f_{monomer} is the fidelity of correct monomer choice within each state. The triple $(I_{\text{struct}}^{\text{pop}}, I_{\text{struct}}^{\text{chan}}, (f_{\text{phase}}, f_{\text{monomer}}))$ is the apparatus signature, and mode classification uses the joint signature rather than any single coordinate.

We distinguish realized population MI from *substrate capacity* $C_S(L) = \sup_{\Pi_X} I_{\text{struct}}^{\text{pop}}(X; Y)$, a property of the substrate class rather than of any specific population. C_S is what enters the principle's (R1) condition; $I_{\text{struct}}^{\text{pop}}$ is what the apparatus measures on a particular biological system. A claim that X is a structurally specific template requires two controls. The bulk-matched control X_{bulk} produces an output Y_{bulk} with k th-order matched marginals but no position-specific information; the structure-scrambled control Y_{scram} has the same composition as Y with positional pattern shuffled. Real templating must give $I_{\text{struct}}^{\text{pop}}(X; Y) \gg I_{\text{struct}}^{\text{pop}}(X; Y_{\text{bulk}})$ and $I_{\text{struct}}^{\text{pop}}(X; Y) \gg I_{\text{struct}}^{\text{pop}}(X; Y_{\text{scram}})$ to a stated significance threshold.

We validate the apparatus on the canonical case. A simulated Watson-Crick replication system with template length L , alphabet size 4, and per-position misincorporation rate ε should give $I_{\text{struct}}^{\text{pop}} = L \cdot$

$[2 - h_b(\varepsilon) - \varepsilon \log_2 3]$. Across 30 sweep cells, the empirical $I_{\text{struct}}^{\text{pop}}$ matches theory to within 0.6% (Figure 1A). The bulk-matched control discriminates cleanly: across 24 non-uniform sweep cells, separation ratios between genuine templating and bulk-matched control range $450\times$ to $1350\times$ (Figure 1B). The signature is robust to the choice of comparison ensemble for $I_{\text{struct}}^{\text{chan}}$: across 10 random four-channel ensembles drawn from a 10-channel pool, every channel’s mode classification was stable.

B. Mode-specific scaling laws

We tested Corollary 1’s mode-specific capacity predictions directly. A two-state cyclic active site, sweeping $N \in \{2, 3, 4, 5, 6, 8, 10\}$ with low-noise channel ($\varepsilon = 10^{-3}$), should approach $I_{\text{struct}}^{\text{pop}} = \log_2 N$ from below. Empirical measurements approach the bound to within estimator precision: $I_{\text{struct}}^{\text{pop}} = 1.0000$ bits at $N = 2$, 2.3219 bits at $N = 5$, 3.3219 bits at $N = 10$ (Figure 2A). For each N , $I_{\text{struct}}^{\text{pop}}$ is constant across L from $L = N$ to $L = 50N$ to within 0.2% (worst case at $N = 2$, $L = 2$; the bound is $< 0.01\%$ for $N \geq 3$). Mode 3 information saturates while Mode 1 information scales linearly with L (Figure 2B), confirming the extensibility-failure prediction within finite-sample noise.

The autocorrelation of Mode 3 output peaks at integer multiples of N , with peak value matching the noise-model prediction $(1 - \varepsilon)^2 + \varepsilon^2/(N - 1)$ (Figure 2C). A Mode 5 simulation sweeping module count $N \in \{2, 4, 8, 16, 32\}$ at alphabet sizes 4 and 20 confirms linear scaling in N , with a sharp length limit: positions beyond the module count carry zero information about the template (Figure 2D). For Mode 2, we simulated the standard genetic code applied to uniformly random DNA templates of L codons. The empirical $I_{\text{struct}}^{\text{pop}}$ matches the analytical value $L \cdot H(\nu) = 4.218L$ bits within finite-sample noise (4.214 at $L = 64$, $n = 5000$). The Mode 1-vs-Mode 2 ratio at matched DNA length is $6/4.218 \approx 1.42$. A nested two-level evolution simulation, with selection on phenotype and the codon-to-amino-acid lookup table itself encoded as a Mode 1 substrate, produced the predicted dynamic: 500 generations drove population fitness from 5% to 43%, with code fidelity converging to 52% as selection acted on phenotype while the recursion acted on the gene encoding the code. The simulation illustrates the mechanism by which Mode 2 inherits parasitically through Mode 1; it does not by itself confirm the biological claim about translation, which rests on independently established molecular biology.

C. Quantitative anchoring to the Drt3 system

The Drt3 system [5] provides a direct biological test. The complex contains two reverse transcriptases and a noncoding RNA. Drt3a uses a conserved ACACAC region of the ncRNA as a Watson-Crick template (Mode 1)

to produce poly(GT) DNA. Drt3b synthesizes the complementary poly(AC) strand without any nucleic acid template, using a cyclic active site in which two conserved gating residues impose alternating state-specific nucleotide selectivity: Glu26 gatekeeps dA selection through hydrogen-bonding to the N^6 amine of dA, and Arg253 stabilizes dA through a cation- π interaction while making three hydrogen bonds with the Watson-Crick edge of dC at the C-state. The Glu26-to-Gln point mutation perturbs dA gating selectively, and the in vitro mutant produces alternating poly(AC) with approximately 80% dA / 20% dG misincorporation at the dA-selecting position while preserving dC selectivity [5, Fig. 4J]. The same protein fold appears in AbiK with different active-site residues, where the system produces random ssDNA [5]. We model Drt3b as a cyclic active-site templating channel; whether the gating residues undergo discrete physical flipping in the structural-biology sense is a separate question that the apparatus-level classification does not require.

We classify these systems by parameterizing Markov-chain channels from the published biochemistry and asking whether the apparatus recovers the expected classifications. Throughout, we label each claim as observed (from Deng *et al.* [5]), derived (analytical consequence), simulated (Markov-chain apparatus on parameterized channels), or predicted (untested experimental target); Table IV maps every claim to its category.

Drt3b WT, parameterized from the (AC) alternating product, gives $I_{\text{struct}}^{\text{pop}} = 1.0000$ bits, saturating the cyclic-channel capacity bound $\log_2 2 = 1$ bit. Phase fidelity $f_{\text{phase}} \approx 0.99$ (cycle alternation intact) and monomer fidelity $f_{\text{monomer}} \approx 0.99$ at both states. Periodicity peaks at lag 2 with peak autocorrelation 0.9802; output marginal frequencies are 0.497 dA, 0.497 dC, 0.003 dG, 0.003 dT; the separation ratio against the bulk-matched control ranges $234\times$ to $1031\times$.

Drt3b Glu26-to-Gln, parameterized with the paper’s 80%/20% dA/dG split at the dA-selecting state, gives $I_{\text{struct}}^{\text{pop}} = 1.0000$ bits (capacity bound preserved) and $f_{\text{phase}} \approx 0.99$ (cycle architecture preserved), but $f_{\text{monomer}} \approx 0.80$ at the A-state with C-state selectivity unchanged. The marginal G fraction rises to 0.102, matching the analytical prediction $0.5 \times 0.20 = 0.10$ for half-cycle dG misincorporation (Deng *et al.* report 0.1016); the periodicity peak drops to 0.83; the separation ratio against the bulk-matched control falls to $\sim 100\times$ (Figure 3A,B). The apparatus classifies the mutant as a cyclic channel with degraded monomer selectivity at one state but intact phase architecture, a distinction that motivates the architectural-versus-selectivity gate classification below.

AbiK same-fold non-templating, parameterized from the published random-product description, gives $I_{\text{struct}}^{\text{pop}} \approx 0.03$ bits and $f_{\text{monomer}} \approx 0.25$ (random over four nucleotides). The apparatus correctly excludes AbiK from the cyclic-templating class and from any structurally specific template class.

Drt3a (Mode 1) and Drt3b (Mode 3), treated as distinct channels in a five-channel comparison ensemble, separate by $\Delta I_{\text{struct}}^{\text{chan}} = 1.0$ bit despite producing nearly identical alternating output (Figure 3C). The channel-as- X observable distinguishes mechanism even when output statistics match: Drt3a’s f_{monomer} characterizes Watson-Crick fidelity against an external ACACAC template, while Drt3b’s f_{phase} and f_{monomer} characterize the cyclic active-site channel. The numerical separation is ensemble-dependent, but the joint signature’s qualitative discrimination is stable across ensemble choices, as the robustness check above shows. WT Drt3b and the Glu26-to-Gln mutant both saturate $I_{\text{struct}}^{\text{pop}}$ at the 1-bit bound, yet differ sharply in marginal G fraction (0.003 versus 0.102) and periodicity peak (0.98 versus 0.83). The mutant and AbiK both produce non-WT compositions but differ in $I_{\text{struct}}^{\text{pop}}$ (1.0 versus 0.03) and f_{monomer} (0.99 versus 0.25). Collapsing the joint signature to a single score discards the information that distinguishes the systems.

D. Family-level application across 1,232 homologs

We applied the apparatus across 1,232 Drt3b homologs identified from the Deng et al. supplementary alignment. Each homolog was assigned to a clade by primary-gate residue (Glu26, Gln26, Asp26, other) and secondary-residue identity at six conserved positions identified through structural alignment. The residue partition follows the Deng paper’s structural assignments: Arg253 (universally invariant) is the architectural gate, enforcing cycle phase via cation- π stabilization of the templating dC; Gly248 (essentially invariant) is a selectivity gate at the C-state via steric exclusion of dG; Glu26 is a primary selectivity gate at the A-state, mimicking a templating nucleobase via two hydrogen bonds with the N⁶ amine of incoming dA [5]. The apparatus makes two predictions. Clades with WT-like primary and secondary residues should fall in-envelope at the WT-Drt3b signature ($I_{\text{struct}}^{\text{pop}} = 1.0$ bit, marginal G fraction < 0.005). Clades with the Glu26-to-Asp primary-gate substitution shorten the carboxylate side chain and degrade A-state selectivity, so the framework predicts elevated dG misincorporation at the A-state with the cycle architecture preserved (Arg253 is invariant in all six such clades).

Of 33 secondary-residue clades assayed, all 33 fall in-envelope (Figure 4A). Of the 6 Glu26-to-Asp primary-gate clades (C06, C14, C20, C27, C28, C42), all 6 show the predicted shift: the marginal G fraction rises from ~ 0.003 in WT-like clades to ~ 0.052 , a 15.7-fold elevation in dG misincorporation at the A-state, consistent with the residual hydrogen-bonding flexibility allowed by the shortened Asp side chain (Figure 4B). The periodicity peak drops from ~ 0.98 to ~ 0.857 , the arithmetic consequence of A-state selectivity loss rather than an independent architectural failure. With state-A residency probabilities (0.85, 0.10, 0.025, 0.025) over (A, G, C, T) and state-C unchanged at 0.99 for C, the analytical period-

2 self-match probability is $\frac{1}{2} \sum_i p_A(i)^2 + \frac{1}{2} \sum_i p_C(i)^2 \approx 0.857$, matching the simulation to three decimal places. The cycle architecture is intact in all six clades (Arg253 invariant), and the apparatus localizes the failure to the A-state selectivity channel rather than the cycle’s phase structure. The application is consistent with the cyclic-templating classification holding across the Drt3b family rather than only at the reference sequence; it is an apparatus-on-parameterized-channels result, not a direct measurement of homolog products, which Deng et al. do not deposit.

E. Universal-gate site-directed-mutagenesis predictions

The architectural-versus-selectivity distinction translates into structurally distinct site-directed mutagenesis predictions, with the family-level Glu26-to-Asp data already providing an empirical anchor on the selectivity side. Arg253 is an architectural gate: its substitution disrupts the cation- π stabilization that holds the cycle in its alternating phase, predicted to collapse $I_{\text{struct}}^{\text{pop}}$ to near zero with the periodicity peak dropping to ~ 0.43 . Gly248 is a selectivity gate at the C-state: its substitution alters monomer selectivity within an intact cycle, predicted to preserve $I_{\text{struct}}^{\text{pop}}$ near WT while elevating marginal G to ~ 0.15 . Glu26 is a selectivity gate at the A-state, structurally analogous to Gly248 but on the complementary half-cycle; the family-level Glu26-to-Asp data confirm the selectivity-gate signature empirically (marginal G ~ 0.052 , cycle phase preserved). Arg253-to-Ala is therefore the cleanest single-experiment test the framework offers, with $100\times$ predicted separation from WT $I_{\text{struct}}^{\text{pop}}$.

Arg253-to-Ala is predicted to collapse $I_{\text{struct}}^{\text{pop}}$ to 0.0090 bits (from WT 0.9997 bits), with the marginal G fraction rising to 0.1265 and the periodicity peak dropping to 0.4325; the cycle architecture is destroyed and the system reduces to compositional bias only. An observed $I_{\text{struct}}^{\text{pop}} > 0.5$ bits would refute the claim that Arg253 is the architectural gate. Gly248-to-Ala is predicted to preserve $I_{\text{struct}}^{\text{pop}}$ at 0.9966 bits, with the marginal G fraction rising to 0.1516 and the periodicity peak dropping to 0.7469; the cycle stays intact and the dG misincorporation comes from a selectivity shift at one state. An observed $I_{\text{struct}}^{\text{pop}} < 0.5$ bits would make Gly248 architectural and the gate-class distinction would fail. Predictions for partial substitutions (Arg253-to-Lys, Arg253-to-His, Gly248-to-Val, Gly248-to-Asp) and the double mutant (Arg253-to-Ala + Gly248-to-Ala) are stated in Table V. The double mutant is the strongest combined test of the gate-class ordering: under the framework’s prediction the architectural collapse should dominate the joint signature, whereas a dominant selectivity signature would show the ordering to be wrong.

F. Finite-population dynamics: bounded-heredity plateaus

We tested Proposition 1 in a pre-specified five-mechanism characterization¹ spanning the inheritance landscape from M0 (no recursion: fails R3 and R4) to M4 (canonical Mode 1: satisfies R1–R4 within R5 scope). M0 is stateless, with phenotype freshly drawn each generation. M1 is lineage-level fixation: at lineage founding a random template is drawn, all lineage members share it, and reproduction copies the lineage label with no individual-level drift (satisfies R3 at the lineage level, fails R4). M2 is lineage-level fixation with re-draw rate $r \in [0, 1]$ (R3 at the lineage level, partial R4 via wholesale re-draw rather than local drift). M3 is individual-level exact copy without drift (R3 at the individual level, fails R4). M4 is individual-level copy with local drift (R3 and R4 at the individual level).

Plateau heights at generation 1000, averaged across 30 replicates with target length $L = 32$, $K = 400$ agents, and selection sharpness $\beta = 10$: M0 = 0.249 (chance), M1 = 0.472, M3 = 0.498, M2 at $r = 0.10 = 0.533$, M4 = 0.972 (Figure 5A; Table VI). Only M4 reaches near-target fitness in the tested model class. M1 and M3 plateau at the maximum fitness present in the initial population’s lineage repertoire, the (R4) failure producing no new drift. M2 has a regime-invariant sweet spot at $r = 0.10$: too little re-draw and the system cannot escape sub-optimal initial draws, too much and the lineage advantage washes out. The sweet spot is invariant within 0.05 across $\beta \in \{2, 5, 10, 20\}$ and $L \in \{16, 32, 64, 128\}$.

A direct head-to-head test confirms the proposition’s content: lineage-level re-draws and individual-level local drift are not interchangeable input sources in the tested model. M2 at $r^* = 0.10$ introduces 2.40 expected new positions per offspring; M4 at $\mu^* = 0.001$ introduces 0.02. M2’s drift rate is $100\times$ M4’s, yet M2 reaches plateau 0.548 while M4 reaches 0.997, and in direct competition at balanced start M4 wins 67%, M2 wins 33%, with no ties. In the tested landscape, targeted local drift converts input into fitness gain more efficiently than wholesale lineage re-draw, so the outcome turns on the structural form of the variation rather than its rate. The qualitative claim survives at long horizons. At 5000 generations with equal-start $K_{M1} = K_{M4} = 200$, M4 wins outright in 24/30 replicates against M1, the remaining 6/30 reflecting cases where M1 fixes early before M4’s drift-and-selection cycle produces a winning lineage; the limited-heredity plateau is durable under the tested update rules,

not a transient. At more asymmetric initial conditions ($K_{M1} = 80$, $K_{M4} = 320$ or $K_{M1} = 20$, $K_{M4} = 380$), M4 wins 30/30.

The proposition’s prediction for the eukaryotic ciliate cortex is consistent with observation (Figure 6). The cortex is built each generation from genome-encoded proteins (re-draw at every cell division) while the existing pattern biases the assembly (lineage-level fixation), operationally close to M2 with low re-draw rate, exactly the regime where the framework predicts limited but heritable variation. Cortical inheritance is documented to propagate non-genomic structural variation for hundreds of generations [3], consistent with M2’s plateau persistence; the cortex does not provide an independently extensible sequence-like substrate, consistent with M2’s bounded ceiling.

G. What the framework establishes

The framework gives biology’s templating chemistries a structural specification they have lacked. The four conditions (R1)–(R4) are properties of the template–catalyst pair under the scope of (R5), not of the population-level dynamics the catalysis hosts. Capacity and generation are separable structural requirements: capacity (R1, R2, R3) bounds the Shannon information the reaction can transmit per round, and generation (R4) controls whether iterated runs can branch off the deterministic closure of a finite initial repertoire. Keeping them separate clarifies which biological chemistries fail what. Drt3b (Mode 3) fails capacity at R2, its cycle having bounded independently variable positions, so drift status is moot once capacity is capped. Prions (Mode 4) fail capacity at both R1 and R2. Translation (Mode 2) recovers capacity parasitically through Mode 1 because its own product fails R3. Population-level kinetics (selection, fitness landscapes, error catastrophe, evolutionary plateaus, lineage formation) are downstream of which structural conditions the underlying chemistry satisfies. Re-grounding the analysis at the catalytic-reaction level makes the operative question which templating chemistries can host biology’s recursion, a property of the reaction rather than of the populations or fitness landscapes it later generates.

The structural conditions are unambiguous; their operationalization is descriptor-relative. The conditions are stated at the kernel level, but applying them to a particular reaction requires a descriptor choice (φ_X, φ_Y) that fixes what counts as a recognized position and what counts as a distinguishable monomer, and the apparatus inherits this descriptor-relativity. This is not a weakness: a descriptor-free taxonomy would be either trivially true (all biology is templated) or untestable (no apparatus could be defined). The framework’s commitment is that, given an admissible descriptor choice, the conditions generate stable mode classifications across reasonable descriptor variations, which we have demonstrated

¹ Analysis plans for each test cohort were written in dispatch markdown files (`dispatch1.md` through `dispatches_v4/`) and reproduced verbatim in the docstring header of each test script before the corresponding result CSVs were generated. The full deposit is archived at 10.5281/zenodo.20060972 (Phase 0 retroactively committing the dispatches; Phases 1–5 adding scripts and result CSVs).

for Drt3b across descriptor choices spanning the natural range.

The framework’s relationship to natural selection is one of layered description. Selection is real as a statement of which catalytic kernels persist in finite populations under environmental filtering, but it is not a force acting inside the kernel itself. The reaction runs because the active-site free-energy landscape, the reactant pool, and the chemical driving allow it to. The kernel produces certain configurations and not others, and those it fails to produce, or produces only unstably, do not carry forward. Selection-talk is descriptive vocabulary applied to the population-level distribution of catalytic kernels, a deflationary reading of selection’s role inside the reaction rather than a denial of its population-level reality.

H. Relation to prior traditions

The framework is complementary to the autopoietic-closure tradition. Rosen [18], Maturana and Varela [19], Pattee [20], and Hofmeyr [21] ask how a living system produces its own templates, the catalysts that read those templates, and the metabolic cycles that produce both. The present framework specifies which classes of templating chemistry can serve as the inheritance carrier in such a self-producing system, identifying Modes 1 and 2 as the only attested candidates. We do not address the origin or self-maintenance of the catalytic machinery itself.

The strongest empirical challenge in the prior literature is yeast prion biology [22, 23]. Prion conformations are real, ecologically prevalent, and evolutionarily active in wild yeast populations, generating heritable phenotypic variation that selection acts on. The framework does not deny prion inheritance; it locates the inheritance carrier on Mode 1. What is transmitted prion-to-prion is a conformational attractor on a polypeptide whose primary sequence is genome-encoded. The prion can stabilize or accelerate adaptation by exposing cryptic genetic variation, but the carrier of heritable distinguishability is the gene encoding the prion-prone protein, itself a Mode 1 substrate. Prions are bounded conformational templating chemistries (Mode 4) supplementing an underlying Mode 1 reaction.

The strongest formal challenge is collectively autocatalytic sets of the GARD/compositional-inheritance class [14, 15]. Vasas *et al.* [16] showed, using the GARD kinetics directly, that compositional inheritance fails the fidelity requirements for open-ended evolution: high-fidelity transmission of composition across cell divisions requires a lognormal distribution of catalytic rate constants so narrow as to be biologically unreachable. The present framework reaches the same verdict by a different route. GARD-class compositional inheritance fails capacity-theorem R2, because composition vectors lack independently variable positions that scale with reaction extent, and fails generation-theorem R4, because fluctuation in composition does not propagate as drift in a rec-

ognized alphabet under a per-site catalytic kernel (there is no per-site recognition step). The two arguments are independent: Vasas *et al.*’s is a dynamical-kinetic-fidelity result for a specific kinetic scheme, ours a structural argument from the catalytic-kernel geometry. They reach the same conclusion, and the present framework supplies the chemical-physics reason behind the empirical failure.

Mode 6 (the ciliate cortex) is the framework’s positive prediction for a 2D-surface templating chemistry. The cortex has been documented since 1965 as a non-genomic structural inheritance system, with grafted patches of inverted basal-body rows propagating as inverted for hundreds of cell generations without genomic change [3, 4]. The framework places it as a bounded-heredity supplementary chemistry dependent on Mode 1: the cortex satisfies R1 and R2 at the surface level (the lattice has multiple unit-types at independently variable positions) but fails R3 and R4 (the surface does not autonomously self-catalyze its own copying with stochastic drift independent of the genome). Pattern transmission depends on Mode 1 acting on the genome to produce the cortical proteins, and pattern drift independent of genome drift cannot accumulate as heritable variation in the recursive sense.

I. Limitations and scope

The biological inventory of six modes is provisional, and new biology may force additions. The taxonomy is descriptive: the modes are derived from attested examples rather than from a deductive enumeration of all possible chemistries satisfying or partially satisfying (R1)–(R4). A more principled future taxonomy would derive modes from active-site dimensionality, recognition cardinality, and recursion topology as orthogonal axes within the framework. The predictions have been tested in coarse-grained Markov-chain models of the catalytic kernel; an atomistic free-energy calculation of Drt3b’s active-site cycle would test whether the two-state picture survives realistic energy-landscape geometry. The DRT family classifications rest on published mechanism descriptions rather than direct apparatus application to raw product reads, and access to deposited Drt3-product sequencing would close that gap.

J. Orthogonal preconditions of life as we know it

The framework specifies the necessary structural conditions on a template–catalyst pair for it to support open-ended Darwinian inheritance. This is one of several essential preconditions for life as we know it. Three others are orthogonal to it. Metabolism, the coupled catalytic network that maintains the cell far from equilibrium, is not supplied by the framework: a population of inheritance-capable substrates without metabolic closure is a chemistry experiment rather than a living

lineage. Compartmentalization lies outside it as well, since an inheritance reaction in bulk solution disperses its descendants into the medium before a lineage can form. Sustained chemical driving (NTP hydrolysis, redox gradients, photochemical input) is a third: without it the catalytic kernel runs once and stops, and (R3) and (R4) collapse to detailed-balance equilibria. Each has its own theoretical literature: Pross [17], Maturana and Varela [19], Hofmeyr [21] for closure-tradition treatments of metabolism, the protocell and vesicle-encapsulation literature for compartmentalization, and nonequilibrium statistical mechanics for chemical driving. Among the substrate-level conditions for the informational spine of biological inheritance, however, the framework is, to our knowledge, the one not previously stated in unified form. Prior work supplied individual conditions or pairs (Schrödinger’s aperiodic crystal, Eigen’s error threshold, Wagner’s neutral networks, Hull’s replicator ontology, Hofmeyr’s (F, A) -systems, Vasas et al.’s GARD refutation; Supplementary §5.8 traces the correspondence), but the capacity/generation separation under a single (R5) scope is new.

IV. CONCLUSIONS

Biology’s templating chemistries sort into a small inventory keyed to the chemical character of the catalytic kernel. The conditions that separate open-ended from bounded inheritance are properties of the template–catalyst pair, and they split into a capacity requirement and a generation requirement with different chemical content: capacity is present when the architecture can store and re-read its configurations, while generation requires, in addition, drift in the recognized alphabet. DNA replication is the only attested chemistry that meets every condition. Translation, the modular biosyntheses, and the ciliate cortex inherit parasitically through DNA because their own products are not re-read; prions and Drt3b are bounded at the substrate level by a small conformational alphabet and a two-state cycle, respectively.

The framework anchors quantitatively to the Drt3 reverse-transcriptase system. It predicts 10% marginal dG misincorporation for the Glu26-to-Gln mutant against a published 10.16%, and it predicts and recovers a 15.7-fold dG elevation across six natural Glu26-to-Asp clades among 1,232 homologs. Two single-residue substitutions at universal gate residues would test its central claim about gate-class architecture: Arg253-to-Ala should collapse the cyclic architecture, and Gly248-to-Ala should produce selectivity loss without architectural collapse. A result discordant with either prediction would refute that claim. Metabolism, compartmentalization, and sustained chemical driving remain orthogonal preconditions for life with their own theoretical lineages. Whether the structural specification holds across novel chemistries is an empirical question: the family analysis, the placement of seven additional DRT systems,

the Mode 6 reframing, and the population-dynamics characterization make the principle a bet that future templating-chemistry findings will decide, rather than a stipulation.

V. MATERIALS AND METHODS

A. Mathematical setup

A templating event is the tuple $(X, \mathcal{O}, S, \Delta G; \varphi_X, \varphi_Y) \rightarrow Y$ where X is the template, $\mathcal{O}$ is the operator (separable molecular machinery, possibly empty or coincident with X), S is the substrate pool, ΔG is the free-energy budget, and φ_X, φ_Y are descriptor maps on template and product spaces. When $\varphi_X = \varphi_Y$ the templating is same-descriptor; otherwise it is cross-descriptor and $\mathcal{O}$ implements the coupling.

The substrate recursion conditions (R1)–(R5), Definitions 1–2, Theorems 1–2, Corollary 1, and Proposition 1 are formalized in Supplementary Mathematical Appendix §5. The diagnostic apparatus is the triple $(I_{\text{struct}}^{\text{pop}}, I_{\text{struct}}^{\text{chan}}, (f_{\text{phase}}, f_{\text{monomer}}))$ defined formally in Supplementary Mathematical Appendix §1. We summarize the operational definitions used in the main paper. Population mutual information $I_{\text{struct}}^{\text{pop}} = H(\varphi_Y(Y)) - \mathbb{E}_X[H(\varphi_Y(Y) | X)]$ is estimated via plug-in entropy estimators on empirical joint distributions. Channel mutual information $I_{\text{struct}}^{\text{chan}}(\mathcal{C})$ is estimated as the mixture-MI under a stated comparison ensemble using $n_{\text{samples}} \geq 5000$ per channel. Per-base fidelities f_{phase} (cycle-state alternation) and f_{monomer} (canonical-monomer selection within a state) are estimated as empirical correctness rates against canonical-output functions. The substrate capacity $C_S(L) = \sup_{\Pi_X} I(X; X')$ is computed analytically per mode (Appendix §2) rather than estimated.

The bulk-matched control $X_{\text{bulk}}, Y_{\text{bulk}}$ preserves k th-order statistics of Y . For $k = 1$, Y_{bulk} is drawn iid from the empirical first-order composition. For $k = 2$, Y_{bulk} is drawn from the maximum-entropy first-order Markov chain preserving empirical nearest-neighbor frequencies. The structure-scrambled control Y_{scram} has the same composition as Y with positional pattern shuffled.

B. Simulation framework

Simulations were implemented in Python 3.10 using NumPy 1.24 and Matplotlib 3.7. Random seeds were set explicitly for reproducibility (per-replicate seed = $42 + \text{cell_idx} \times 100 + \text{rep_idx}$). Computations ran on a 64-CPU compute node. Code and raw simulation outputs are available at 10.5281/zenodo.20272479 (full submission-state snapshot); see Data Availability for the relationship to the earlier pre-specification deposit at 10.5281/zenodo.20060972.

C. Test A.1: Mode 1 length-scaling validation

Templates X of length L over alphabet $\{A, C, G, T\}$ drawn uniformly. Products Y generated by Watson-Crick channel with per-position misincorporation rate ε . Sweep $\varepsilon \in \{0.001, 0.01, 0.05, 0.10, 0.25\}$ and $L \in \{10, 25, 50, 100, 200, 500\}$ with $n = 5000$ per cell. Per-position MI $I(X_i; Y_i)$ computed by plug-in estimator. Theoretical comparison: $I_{\text{struct}}^{\text{pop}} = L \cdot [2 - h_b(\varepsilon) - \varepsilon \log_2 3]$.

D. Test A.2: bulk-matched control discrimination

Three template biases tested (π_{uniform} , $\pi_{\text{AT-skew}}$, $\pi_{\text{GC-skew}}$). Bulk control draws Y_{bulk} iid from analytical output marginal Q at each position, with X_{bulk} uncorrelated. Pass: $I_{\text{struct}}^{\text{pop}} < 0.05$ bits, separation ratio $> 20\times$.

E. Test B: Mode 3 information saturation

Two-state cyclic active site simulation. Sweep $N \in \{2, 3, 4, 5, 6, 8, 10\}$, $L \in \{N, 5N, 10N, 50N\}$, $\varepsilon = 10^{-3}$, $n = 5000$ per cell. $I_{\text{struct}}^{\text{pop}}$ computed against template ensemble equal to phase index. Theoretical bound: $I_{\text{struct}}^{\text{pop}} \leq \log_2 N$.

F. Test C: Mode 5 module-bounded scaling

Modular conveyor channels with N modules, alphabet sizes $|\mathcal{A}| \in \{4, 20\}$. Sweep $N \in \{2, 4, 8, 16, 32\}$, $L = 5N$, $n = 5000$ per cell. Theoretical bound: $I_{\text{struct}}^{\text{pop}} = N \cdot \log_2 |\mathcal{A}|$ for $L \geq N$.

G. Test D: Mode 2 codon-degeneracy capacity

Standard genetic code (NCBI table 1). Random DNA templates length L codons, $L \in \{16, 32, 64, 128\}$, $n = 5000$ per cell. Output measured per amino acid (with stop). Theoretical value: $L \cdot H(\nu) = 4.218L$ bits.

H. Test F: apparatus signature on Drt3 channels

Five-channel ensemble: $\{\text{Drt3a Mode 1, Drt3b Mode 3 WT, Drt3b Glu26-to-Gln, Drt3a homopolymer null, AbiK random}\}$. Channel ensemble robustness: 10 alternative four-channel ensembles drawn from a 10-channel pool. Mode classification stability checked across all ensemble draws.

I. Test G: Drt3b homolog family-level apparatus application

1,232 Drt3b homologs from Deng et al. supplementary alignment. Clade assignment by primary-gate residue (Glu26, Gln26, Asp26, other) and secondary-residue identity at six conserved positions. Apparatus signature predicted per clade; in-envelope assessment at the WT-Drt3b reference signature.

J. Test H: five-mechanism population-dynamics characterization

Mechanisms M0–M4 as defined in Results. Target length $L_{\text{TARGET}} \in \{16, 32, 64, 128\}$. Population size $K = 400$. Selection sharpness $\beta \in \{2, 5, 10, 20\}$. Mutation rate (M4) $\mu \in \{0.001, 0.01, 0.05, 0.10\}$. Re-draw rate (M2) $r \in \{0.01, 0.05, 0.10, 0.25, 0.50\}$. 30 replicates per cell. Plateau height equals mean fitness at gen 1000. Direct competition uses balanced start with equal initial population sizes; the head-to-head winner is the 5000-gen majority. Long-horizon stability assessed at 5000 generations.

K. Test I: universal-gate SDM predictions

Arg253 substitutions: Arg253-to-Ala, Arg253-to-Lys, Arg253-to-His. Gly248 substitutions: Gly248-to-Ala, Gly248-to-Val, Gly248-to-Asp. Double mutant: Arg253-to-Ala + Gly248-to-Ala. Markov-chain channels parameterized from structural homology arguments and the Drt3b WT reference. $L = 64$, 30 replicates, $n_{\text{samples}} = 5000$ per replicate. 95% CIs from beta-binomial.

DATA AVAILABILITY

All simulation code, raw simulation outputs, the diagnostic apparatus implementation, the family-level analysis pipeline, and the dispatch markdown files containing the pre-specified analysis plans for each test cohort are deposited in a permanent public repository (github.com/khatvangi/templating-substrates) and archived on Zenodo with two complementary DOIs [29, 30]. 10.5281/zenodo.20060972 is the *pre-specification anchor* (release v0.0-prereg), archiving the analysis-plan dispatches at the moment of pre-registration before the empirical work began. 10.5281/zenodo.20272479 is the *reproducibility snapshot* (release v1.0-submission), archiving the full submission state including manuscript, supplementary appendix, bibliography, figures, all test scripts, and all result CSVs. Phylogenetic input files (`science_aed1656_data_s4--s8`) are derived from the published supplementary material of Deng *et al.* [5]. The deposit's git history records the commit ordering: Phase

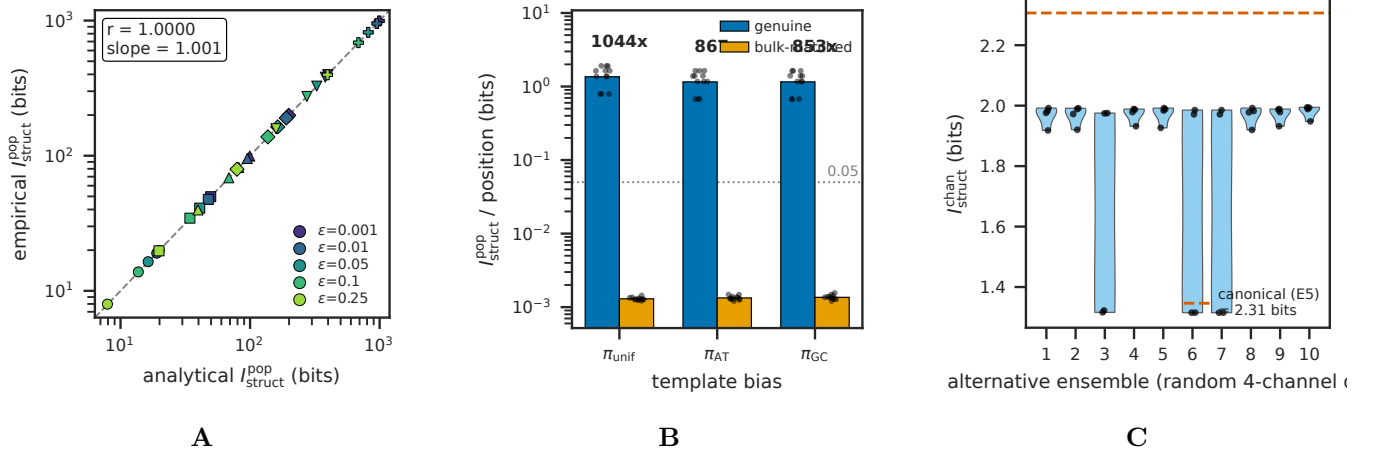
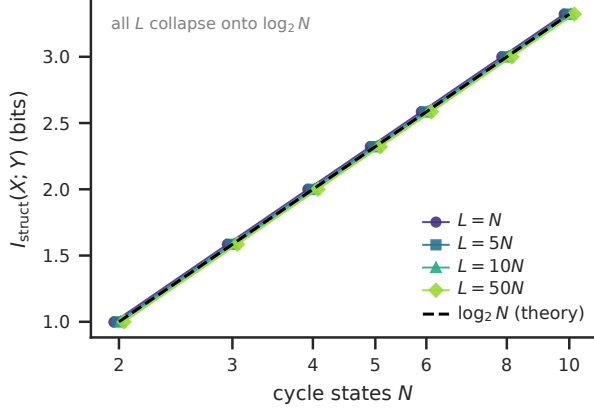


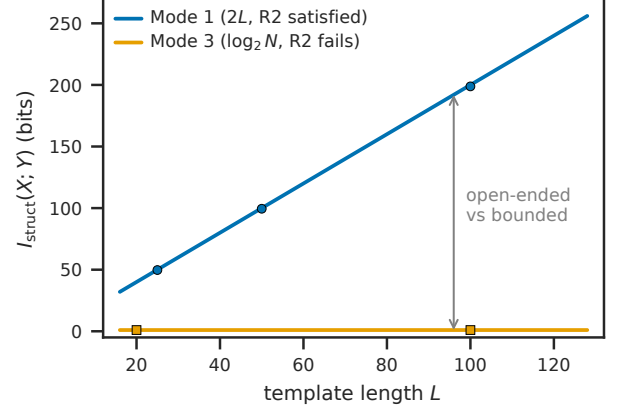
FIG. 1: **Apparatus validation and bulk-matched control.** (A) Empirical $I_{\text{struct}}^{\text{pop}}$ versus analytical prediction across the $\varepsilon \times L$ sweep, showing close tracking of analytical capacity in noise-free Mode 1 templating (test A1). (B) Separation ratio $I_{\text{struct}}^{\text{pop, genuine}} / I_{\text{struct}}^{\text{pop, bulk}}$ across three template-bias conditions, showing $> 20\times$ separation between genuine templating and bulk-matched controls (test A2). (C) Comparison-ensemble robustness: $I_{\text{struct}}^{\text{chan}}$ across 11 alternative comparison ensembles (test G3); the qualitative classification of Drt3a as Mode 1 and Drt3b as Mode 3 is preserved across all ensembles, confirming that the apparatus’s classificatory verdict is stable under reasonable analyst choices.

TABLE I: **Substrate recursion principle: four structural conditions plus one scope-defining condition.**

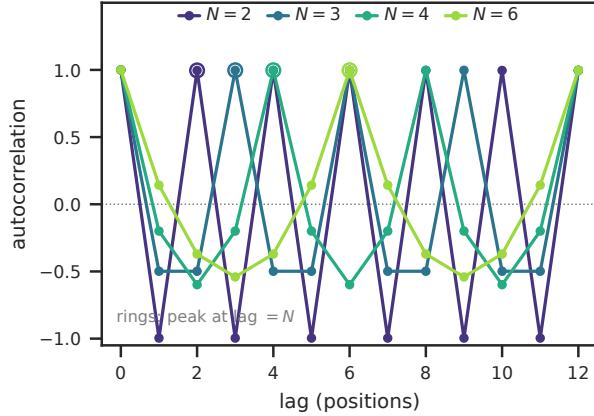
Condition	Name	Statement	Failure mode in biological substrates
R1	Positive entropy per recognized site	At each addressable position, recognition alphabet $ \Sigma \geq 2$ with $h_{\text{site}} = \log_2 \Sigma > 0$	Mode 4 (prions, conformer-state): recognition alphabet is small attractor repertoire, h_{site} bounded at $\log_2 K$ at the macro level
R2	Unbounded independently variable position count	Non-recurrent sites with $\liminf_{L \rightarrow \infty} N(L) = \infty$; substrate is not constrained to a finite recurrent phase repertoire (independence in the first-order descriptor-relative sense, not strong joint-statistical independence)	Mode 3 (cyclic): $N(L)$ bounded at cycle length N regardless of product length. Mode 5 (modular): bounded at module count N . Mode 4: $N(L) \approx 1$ trivially
R3	Same-operation recursion	Output is structurally readable by the same operation, possibly via reversible involution	Mode 2 (translation): protein product is not ribosome-readable; recursion is parasitic on Mode 1. Mode 5 (NRPS): peptide product is not the assembly line. Mode 6 (cortex): surface does not self-copy autonomously
R4	Heritable local drift in recognized alphabet	Drift lies in the recognition alphabet of (R1); content and templating function are the same physical entity. “Drift” rather than “error”: the kernel produces what its physics produces, and the dispersion is a property of the kernel ensemble, not departure from an external target	Crystals (excluded from biology): defect propagation in bounded geometric alphabet, not extensible alphabet. Mode 6: pattern drift independent of genome drift cannot accumulate
R5	Scope: non-teleological operation	Operation is intrinsic physicochemistry, not externally specified; no target, loss function, or agent within the operation. Scope-defining: demarcates domain of (R1)–(R4), does not introduce a structural condition	LLM training (non-biological): operations are externally specified by loss function; gradient updates are not substrate-intrinsic recursion. Outside scope rather than structurally failing



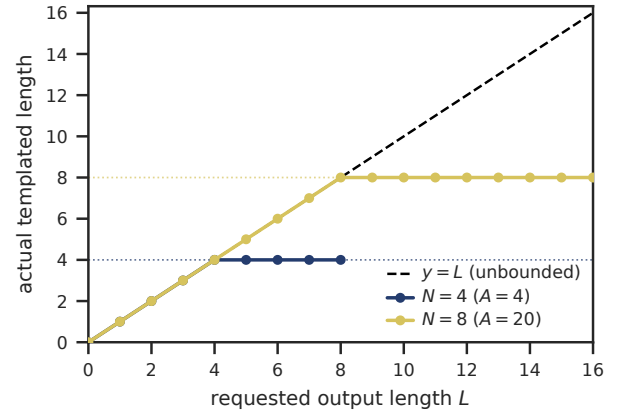
A



B



C



D

FIG. 2: Mode-specific information-capacity scaling. (A) Mode 3 information capacity $I_{\text{struct}}^{\text{pop}}$ saturates at $\log_2 N$ as cycle states N increase (test B), regardless of template length L , confirming that the macro-level capacity bound is set by the cycle's recurrent phase repertoire and not by output length. (B) Mode 1 capacity scales linearly in L (slope $\log_2 |\mathcal{A}| = 2$) while Mode 3 saturates at $\log_2 2 = 1$ bit; the structural difference between (R2)-satisfying and (R2)-failing substrates is visible in this single panel. (C) Mode 3 output autocorrelation peaks at lag = N for $N \in \{2, 3, 4, 6\}$, the canonical signature of cyclic-phase templating. (D) Mode 5 output length is capped at the module count N regardless of requested L (test C), confirming the (R2)-failure-at-module-count diagnosis.

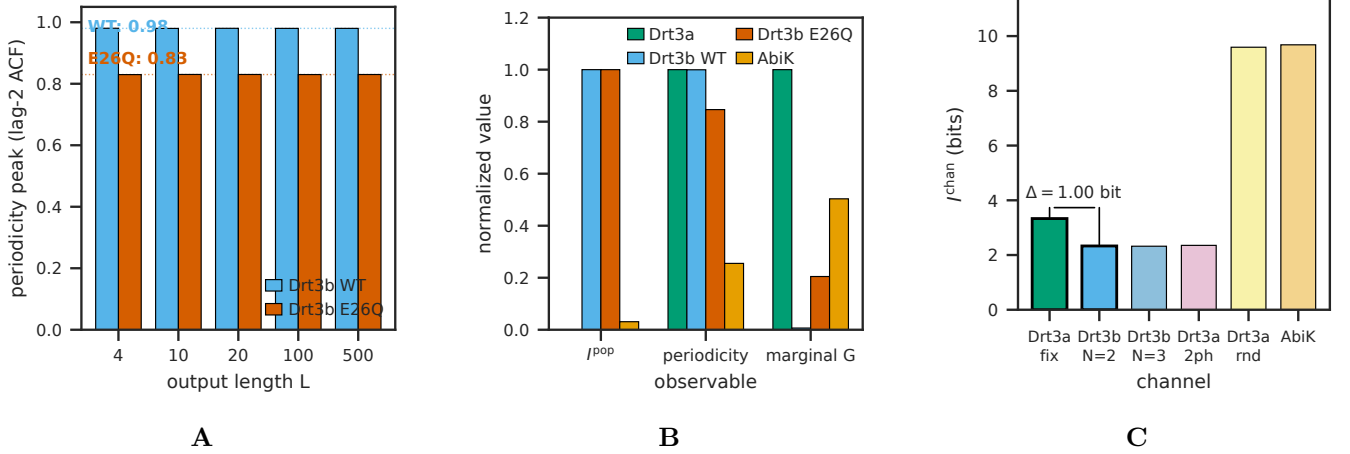


FIG. 3: Apparatus signature on the Drt3 system. (A) Drt3b WT versus Glu26-to-Gln periodicity peaks (test E2): WT shows $f_{\text{phase}} \approx 0.99$, the mutant shows $f_{\text{phase}} \approx 0.83$, confirming the analytical prediction of half-cycle dG misincorporation in the gate-broken mutant ($0.5 \times 0.20 = 0.10$ marginal G fraction; observed 0.1016 in Deng et al. 2026 Fig. 4J). (B) Apparatus triple ($I^{\text{pop}}_{\text{struct}}$, periodicity peak, marginal G fraction) separating WT Drt3b, Glu26-to-Gln, AbiK, and Drt3a in joint-observable space (test G2); the four systems classify cleanly despite producing nearly identical alternating output for some pairs. (C) Channel-mutual-information separation: against the stated five-channel comparison ensemble, the Drt3a Watson-Crick channel gives $I^{\text{chan}}_{\text{struct}} = 3.33$ bits and the Drt3b $N = 2$ channel gives $I^{\text{chan}}_{\text{struct}} = 2.33$ bits, yielding $\Delta I^{\text{chan}}_{\text{struct}} = 1.00$ bit between Mode 1 and Mode 3.

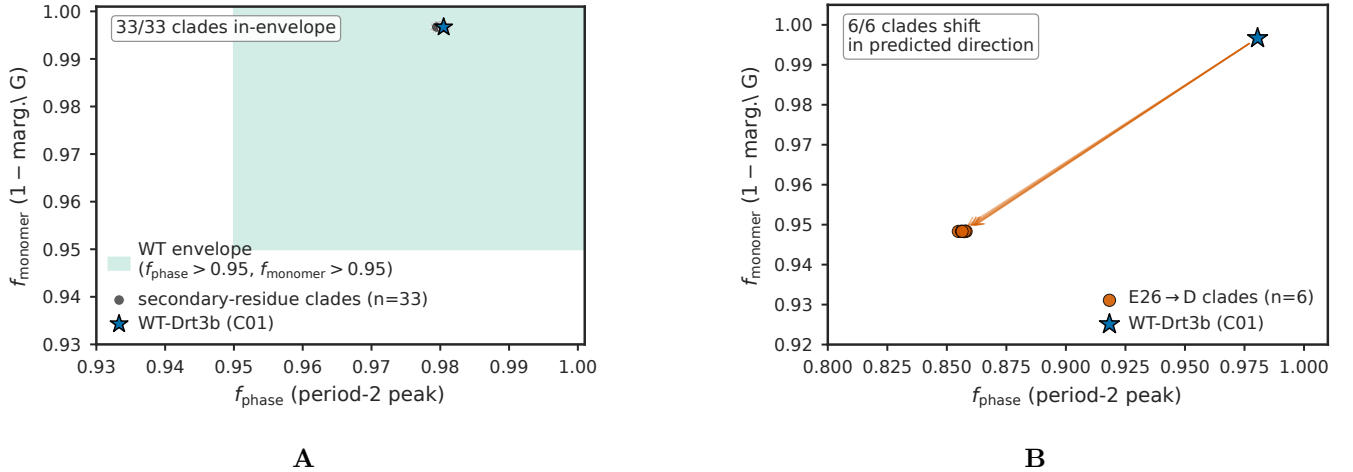
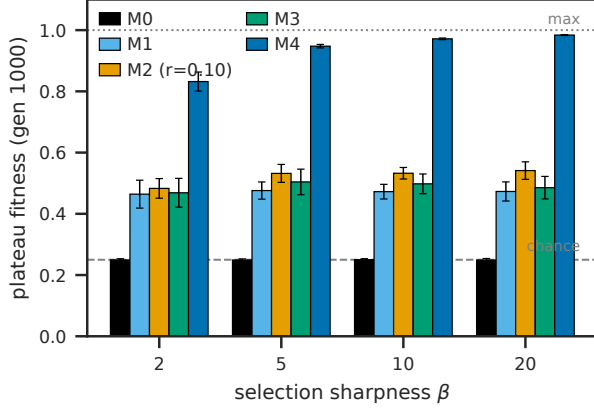
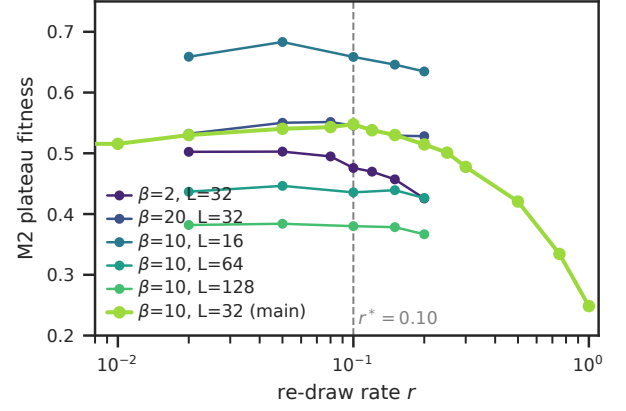


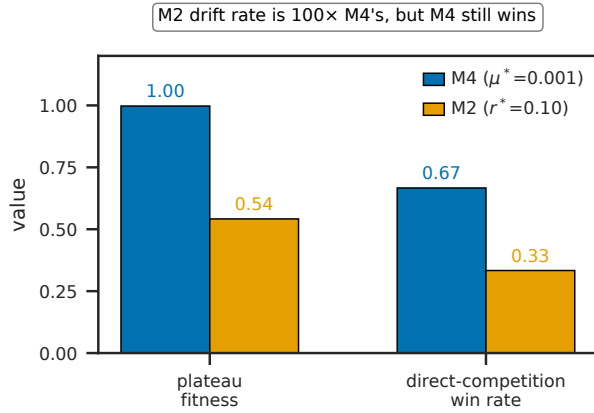
FIG. 4: Drt3b family-level apparatus application across 1,232 homologs. (A) In-envelope behavior of 33 secondary-residue clades (test F): all 33 fall within the WT-Drt3b apparatus envelope, supporting the inference that secondary-residue diversity does not break the cyclic-conformational mode. (B) Glu26-to-Asp selectivity-gate signature in 6 primary-gate clades (C06, C14, C20, C27, C28, C42): the marginal G fraction rises from ~ 0.003 in WT-like clades to ~ 0.052 in the Glu26-to-Asp clades, a 15.7-fold elevation, consistent with residual hydrogen-bonding flexibility from the shortened Asp side chain. The periodicity peak drops to ~ 0.857 as the arithmetic consequence of A-state selectivity loss (analytical $\frac{1}{2} \sum p_A^2 + \frac{1}{2} \sum p_C^2 \approx 0.857$), not as an independent phase failure: cycle architecture is intact (Arg253 invariant in all 6 clades). All 6 clades show the framework-derived selectivity shift, supporting the classification of Glu26 as a primary selectivity gate distinct from the architectural gate Arg253. Apparatus results are computed on Markov channels parameterized from residue assignments at conserved positions, not from raw product sequencing.



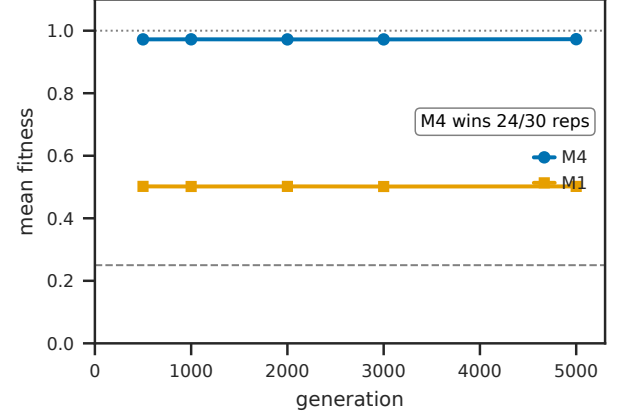
A



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FIG. 5: **Population-dynamic plateaus from substrate condition failure.** (A) Mechanism-specific plateau heights $\bar{F}_{1000}$ across selection sharpness $\beta \in \{2, 5, 10, 20\}$ (test H3): M0 (R3, R4 fail) plateaus at chance (0.250); M1, M3 (R4 fails) plateau near initial-population maximum (~ 0.47 to 0.50); M2 at $r = 0.10$ (partial R4) plateaus at ~ 0.53 ; M4 (R1–R4 satisfied) reaches ~ 0.97 . (B) M2 sweet-spot at $r^* \approx 0.10$, regime-invariant within ± 0.05 across $\beta \in \{2, 5, 10, 20\}$ and $L \in \{16, 32, 64, 128\}$ (test H5). (C) M4 versus M2 head-to-head at matched per-offspring drift rate (test H5): M2 introduces 2.40 new positions/offspring at $r^* = 0.10$; M4 introduces 0.02 at $\mu^* = 0.001$; M4 still wins 67%/33% in direct competition, demonstrating that targeted local drift converts to fitness gain more efficiently than wholesale lineage re-draw. (D) Long-horizon (5000 generations) M4 versus M1 trajectories at equal-start initial conditions ($K_{M1} = K_{M4} = 200$): M4 wins 24/30 replicates; M1's plateau is durable but bounded.

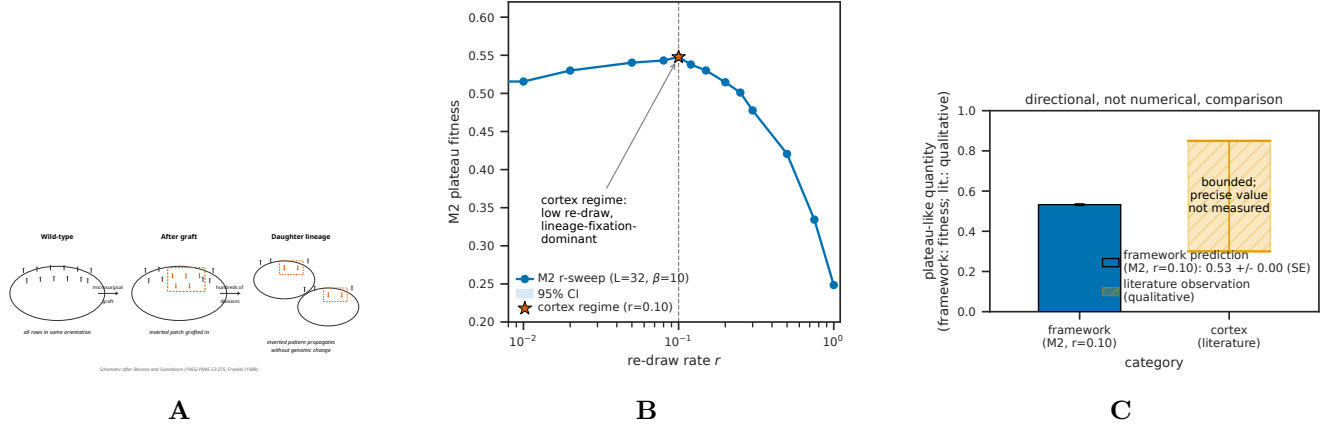


FIG. 6: Mode 6 reframing and ciliate cortex bounded-heredity retrodiction. (A) Schematic of cortical inheritance after Beisson and Sonneborn [3]: a microsururgically grafted patch of inverted ciliary basal-body rows propagates as inverted across hundreds of cell divisions in *Paramecium* without genomic change, demonstrating that the cortex inherits non-genomic structural variation. (B) Operational analog within the M2 mechanism class: at every cell division, cortical units are re-drawn from genome-encoded parts, and the existing pattern biases the orientation of newly assembled units. The framework places the cortex in the M2 regime with $r \approx 0.10$ (low re-draw, lineage-fixation-dominant), consistent with the Beisson-Sonneborn observation. (C) Predicted versus observed: the framework predicts bounded but persistent heredity for Mode 6 substrates (M2 plateau at ~ 0.53 in simulation), and the literature reports bounded but persistent cortical inheritance (hundreds of generations of pattern propagation, no qualitatively new cortical structures accumulated over evolutionary timescales). The comparison is directional, not quantitative: the literature’s bounded heredity is a qualitative observation, not a fitness measurement.

TABLE II: Six modes of biological templating, classified by substrate recursion principle conditions. The Mode 2 entry gives the one-step channel mutual information under uniform codon input; its recursive capacity C_S is near zero because R3 fails.

Mode	Substrate	Conditions failed	Capacity scaling	Copyability	Examples
1	1D linear sequence on separate molecule	None (satisfies R1–R4 within R5 scope)	$L \cdot \log_2 \mathcal{A} $	Auto-catalytic via complementarity (R3 involution)	DNA replication, Drt3a, telomerase, DRT9
2	1D sequence + separable lookup table	R3 (parasitic recursion via Mode 1)	$I_{\text{struct}}^{\text{pop}} \leq L \cdot H(\nu)$ (uniform input); $C_S \approx 0$	Via Mode 1 on Ribosomal translation encoding gene	
3	3D protein cyclic dynamics, N states	with R2 (bounded position count N)	$\leq \log_2 N$ (saturating)	Not auto-catalytically copyable	Drt3b ($N = 2$). DRT7 ($N = 1$) is structural-selectivity boundary, $\log_2 1 = 0$ positional capacity
4	Discrete conformational state	R1 (h_{site} bounded at $O(1)$), R2 ($N \approx 1$)	length-independent	Propagation, not copy	Prions, amyloids, DRT1 filament
5	Spatial sequence of N modules	R2 (bounded at module count), R3 (parasitic)	$N \cdot \log_2 k$, output bounded by N	Via Mode 1+2 on module gene	NRPS, polyketide synthases
6	2D pattern (lattice, surface, cortex)	R3 (no self-copy), R4 (no autonomous heritable drift)	autonomous R1, R2 satisfied at surface; bounded by Mode 1 dependency	Limited heredity, parasitic on Mode 1	Quasicrystal adlayers; ciliate cortex

0 retroactively commits the dispatches with their original filesystem mtimes preserved as commit metadata, and Phases 1–5 commit the test scripts, result CSVs, and figure pipeline. The pre-specified predictions are re-

produced verbatim in the docstring headers of the test scripts committed in Phases 2–4.

TABLE III: DRT family classification under the framework.

System	Mode	Mechanism summary	Reference
DRT1	Mode 4	Filamentous quaternary state with ssDNA effector	[6]
DRT2	Mode 1	Rolling-circle reverse transcription on ncRNA template, <i>neo</i> effector	[24, 25]
DRT3a	Mode 1	Watson-Crick on ACACAC ncRNA region, produces poly(GT)	[5]
DRT3b	Mode 3 ($N = 2$)	Cyclic protein-conformational gating, produces poly(AC)	[5]
DRT4	Out of scope	Random ssDNA, biasing-participant (toxin)	[26]
DRT6	Out of scope	Random ssDNA, structural homolog of DRT4	—
DRT7	Mode 3 boundary ($N = 1$)	Single-state arginine-rich pocket, $\log_2 1 = 0$ positional capacity; structural channel selectivity remains, but no positional templating; classified as boundary case rather than full Mode 3 instance	[7]
DRT9	Mode 1	Poly-U ncRNA template, produces poly(dA) homopolymer	[27]
DRT10	Mode 1	Translocation-based reuse of short ncRNA template; telomerase ancestor	[28]

FUNDING

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

DECLARATION OF COMPETING INTEREST

Declarations of interest: none.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work the author used an AI assistant (Anthropic Claude) for language editing and to improve the clarity and structure of the prose. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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- [1] J. Greenwald, M. P. Friedmann, and R. Riek, *Angewandte Chemie International Edition* **55**, 11609 (2016).
- [2] C. T. Walsh, H. Chen, T. A. Keating, *et al.*, *Current Opinion in Chemical Biology* **5**, 525 (2001).
- [3] J. Beisson and T. M. Sonneborn, *Proceedings of the National Academy of Sciences* **53**, 275 (1965).
- [4] J. Frankel, *Pattern Formation: Ciliate Studies and Models* (Oxford University Press, 1989).
- [5] P. Deng, H. Lee, C. Armijo, H. Wang, A. Gao, *et al.*, *Science* 10.1126/science.aed1656 (2026), first Release, 16 Apr 2026.
- [6] N. Neville, N. V. Johnson, E. E. Escobar, C.-H. Chiang, A. Nreca, S. R. Johnson, N. Dai, A. Hanneman, I. R. Corrêa, Jr., J. S. McLellan, and R. J. Trachman, III, *bioRxiv* 10.64898/2026.03.07.710306 (2026), preprint, posted 9 March 2026.
- [7] X.-Y. Song, Y. Xia, J.-T. Zhang, X.-Y. Wei, H. Qi, L. Li, *et al.*, *bioRxiv* 10.64898/2026.04.18.719102 (2026), preprint, posted 19 April 2026; Song, Xia, and Zhang contributed equally.
- [8] M. Eigen, *Naturwissenschaften* **58**, 465 (1971).
- [9] M. Eigen, J. McCaskill, and P. Schuster, *Advances in Chemical Physics* **75**, 149 (1989).
- [10] E. Szathmáry, *Philosophical Transactions of the Royal Society B* **355**, 1669 (2000).
- [11] D. L. Hull, *Annual Review of Ecology and Systematics* **11**, 311 (1980).
- [12] J. Maynard Smith, *Philosophy of Science* **67**, 177 (2000).
- [13] E. Schrödinger, *What is Life? The Physical Aspect of the Living Cell* (Cambridge University Press, 1944).
- [14] S. A. Kauffman, *Journal of Theoretical Biology* **119**, 1 (1986).
- [15] W. Hordijk, *BioScience* **63**, 877 (2012).
- [16] V. Vasas, E. Szathmáry, and M. Santos, *Proceedings of the National Academy of Sciences* **107**, 1470 (2010).
- [17] A. Pross, *Journal of Systems Chemistry* **2**, 1 (2011).
- [18] R. Rosen, *Bulletin of Mathematical Biophysics* **20**, 245 (1958).
- [19] H. R. Maturana and F. J. Varela, *De Máquinas y Seres Vivos: Autopoiesis – la organización de lo vivo* (Editorial Universitaria, Santiago, 1972).
- [20] H. H. Pattee, in *Towards a Theoretical Biology, Vol. 4*, edited by C. H. Waddington (Edinburgh University Press, 1972) pp. 248–258.
- [21] J.-H. S. Hofmeyr, in *Handbook of Anticipation*, edited by R. Poli (Springer, 2017).

TABLE IV: **Epistemic status of the Drt3 claims.** Results in this paper fall into four categories. *Observed* = published experimental fact from Deng *et al.* [5]. *Derived* = arithmetic or analytical consequence of an observed fact under the framework. *Simulated* = Markov-chain apparatus output on parameters read from the published biochemistry or from structural homology. *Predicted* = framework-derived experimental target not yet measured.

Claim	Status	Source / basis
Drt3b produces alternating poly(AC) without nucleic acid template	Observed	Deng <i>et al.</i> [5] biochemistry
Glu26-to-Gln produces 80%/20% dA/dG split at A-state	Observed	Deng <i>et al.</i> [5, Fig. 4J]
AbiK same-fold produces random ssDNA	Observed	Deng <i>et al.</i> [5]
1,232-homolog alignment with Glu26, Gly248, Arg253 conservation pattern	Observed	Deng <i>et al.</i> [5] Data S3
Marginal G fraction ≈ 0.10 in Glu26-to-Gln (analytical 0.5×0.20)	Derived	Half-cycle dG misincorporation arithmetic
Period-2 self-match ≈ 0.857 in Glu26-to-Asp clades	Derived	$\frac{1}{2} \sum p_A^2 + \frac{1}{2} \sum p_C^2$ from parameterization
Mode 3 capacity ceiling $\log_2 N$	Derived	Capacity theorem applied to N -state cyclic substrate
WT Drt3b: $I_{\text{struct}}^{\text{pop}} = 1.0$ bits, $f_{\text{phase}} \approx 0.99$, separation ratio $234 \times -1031 \times$	Simulated	Apparatus on Markov channel parameterized from observed (AC) output
Glu26-to-Gln: $I_{\text{struct}}^{\text{pop}} = 1.0$ bits, $f_{\text{monomer}} \approx 0.80$ A-state	Simulated	Apparatus on Markov channel parameterized from observed 80%/20% split
33/33 secondary-residue clades in-envelope	Simulated	Apparatus on per-clade Markov parameterizations from residue assignments
15.7-fold elevated dG in 6 Glu26-to-Asp clades	Simulated	Apparatus on per-clade parameterizations; not measured from raw products
Drt3a vs Drt3b separation $\Delta I_{\text{struct}}^{\text{chan}} = 1.0$ bit	Simulated	Channel-MI against stated 5-channel ensemble
Arg253-to-Ala: $I_{\text{struct}}^{\text{pop}} \rightarrow 0.009$, phase peak $\rightarrow 0.43$, cycle collapse	Predicted	SDM experimental target; not yet performed
Gly248-to-Ala: $I_{\text{struct}}^{\text{pop}}$ near WT, marginal G $\rightarrow 0.15$, cycle intact	Predicted	SDM experimental target; not yet performed
Arg253-to-Ala + Gly248-to-Ala double mutant: Arg253-to-Ala signature dominates	Predicted	SDM experimental target; test of gate-class ordering

[22] R. Halfmann and S. Lindquist, *Science* **330**, 629 (2010).
[23] R. Halfmann *et al.*, *Nature* **482**, 363 (2012).
[24] M. E. Wilkinson, L. C. Tang, *et al.*, *Science* 10.1126/science.adq3977 (2024).
[25] S. Tang, V. Conte, D. J. Zhang, *et al.*, *Science* 10.1126/science.adq0876 (2024).
[26] Y. Wang, H. Wu, J. Li, Q. An, Z. Tian, and Z. Deng, *Nature Communications* **17**, 289 (2025), wang and Wu contributed equally; PMID 41330908; PMC PMC12783706.
[27] K. Mou *et al.*, *Science* 10.1126/science.ads4639 (2025).

[28] S. Tang, J. L. Ramirez, M. R. Mestre, D. J. Zhang, M. Wang, T. Wiegand, Y. Ma, M. Jovanovic, R. Pinilla-Redondo, and S. H. Sternberg, *bioRxiv* 10.1101/2025.10.16.682844 (2025), preprint, posted 16 October 2025.
[29] B. Kiran, Templating substrates: reproducibility snapshot (v1.0-submission), [dataset] Zenodo (2026).
[30] B. Kiran, Templating substrates: pre-specification anchor (v0.0-prereg), [dataset] Zenodo (2026).

TABLE V: **Universal-gate site-directed-mutagenesis predictions (architectural versus selectivity).**
Predictions at $L = 64$, 30 reps, $n_{\text{samples}} = 5000$ per rep. 95% CIs from beta-binomial.

Mutant	Class	$I_{\text{struct}}^{\text{pop}}$ (bits) [95% CI]	Marginal CI]	G [95% CI]	Period-2 peak [95% CI]	Test
WT (Drt3b)	—	0.9997 [0.9987, 1.0000]	0.0033 [0.0035]	[0.0031, 0.9802]	[0.9794, —]	—
Arg253-to-Ala	Architectural	0.0090 [0.0065, 0.0120]	0.1265 [0.1274]	[0.1255, 0.4325]	[0.4303, 0.4345]	Observed $I_{\text{struct}}^{\text{pop}} > 0.5$ bits refutes Arg253-as-architectural-gate WT-like result would show Arg253's role is more limited
Arg253-to-Lys	Partial architectural	0.94	0.027	0.85		Intermediate; checks dose-response
Arg253-to-His	Partial architectural	0.85	0.068	0.70		Observed $I_{\text{struct}}^{\text{pop}} < 0.5$ bits makes Gly248 architectural and the gate-class distinction fails
Gly248-to-Ala	Selectivity	0.9966 [0.9936, 0.9991]	0.1516 [0.1528]	[0.1505, 0.7469]	[0.7455, 0.7479]	Confirms cycle preserved
Gly248-to-Val	Selectivity	0.972	0.077	0.82		Confirms cycle preserved
Gly248-to-Asp	Selectivity	0.971	0.102	0.79		A dominant Gly248-to-Ala signature would show the architectural-versus-selectivity ordering is wrong
Arg253-to-Ala + Gly248-to-Ala	Architectural-dominant	0.0084 [0.0059, 0.0116]	0.1519 [0.1537]	[0.1506, 0.3872]	[0.3855, 0.3892]	

TABLE VI: **Inheritance-mechanism plateau heights (Test H3, Corollary 1).** Plateau = mean fitness at gen 1000 averaged over 30 replicates. $K = 400$, $L_{\text{TARGET}} = 32$, $\mu_{M4} = 0.01$. Theoretical maximum 1.0; chance baseline 0.250. Mechanism-condition mapping in column 2.

Mechanism	Conditions failed	$\beta = 2$	$\beta = 5$	$\beta = 10$	$\beta = 20$
M0 (stateless)	R3, R4	0.249	0.249	0.250	0.249
M1 (lineage fixation)	R4 (no individual-level drift)	0.464	0.476	0.472	0.473
M2 ($r = 0.10$)	R4 partial (wholesale re-draw, not local drift)	0.483	0.532	0.533	0.541
M3 (individual exact copy)	R4 (no drift)	0.469	0.504	0.498	0.485
M4 (individual + local drift)	None (R1–R4 satisfied within scope)	0.832	0.948	0.972	0.984

Supplementary Mathematical Appendix

A theory of templating substrates

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This appendix provides formal definitions and proofs for the mathematical claims in the main paper. The material is organized into five sections corresponding to the apparatus, mode-specific information bounds, descriptor-relativity, the formal sense of bulk-matched controls, and the capacity and generation theorems with their bounded-heredity corollary and finite-population proposition.

The appendix is written for a reader with graduate-level information theory ([Cover and Thomas, 2006](#)) and basic familiarity with discrete memoryless channels. Notation follows the main paper; $|\mathcal{A}|$ denotes alphabet cardinality, $h_b(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ is binary entropy, and $D_{\text{KL}}(P \parallel Q)$ is Kullback-Leibler divergence.

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1 Formal apparatus definitions

1.1 The templating event

A templating event is a tuple $\mathcal{E} = (X, O, S, \Delta G; \varphi_X, \varphi_Y, Y)$ where:

- $X \in \mathcal{X}$ is the template, drawn from a template population with distribution Π_X (which may be a point mass δ_x for a single fixed template).
- O is the operator (separable molecular machinery; possibly empty if templating is autocatalytic).
- S is the substrate pool from which Y is constructed.
- ΔG is the free-energy budget driving the templating reaction.
- $\varphi_X : \mathcal{X} \rightarrow \mathcal{X}'$ and $\varphi_Y : \mathcal{Y} \rightarrow \mathcal{Y}'$ are descriptor maps on template and product spaces.
- Y is the structured product, distributed as $Y \mid (X, O, S, \Delta G) \sim P_{\text{chan}}(\cdot \mid X)$, where P_{chan} is the templating channel induced by O acting on $(X, S, \Delta G)$.

The templating channel $P_{\text{chan}}(\cdot \mid X)$ is a stochastic kernel from $\mathcal{X}$ to $\mathcal{Y}$. When $\varphi_X = \varphi_Y$ the templating is *same-descriptor*; otherwise it is *cross-descriptor* and O implements the descriptor coupling.

1.2 Population mutual information

Definition 1 (Population MI). For a templating event $\mathcal{E}$ with template population $\mathcal{X}$ and distribution Π_X ,

$$I_{\text{struct}}^{\text{pop}}(\mathcal{E}) := I(\varphi_X(X); \varphi_Y(Y)) = H(\varphi_Y(Y)) - H(\varphi_Y(Y) \mid \varphi_X(X)),$$

where the marginal $H(\varphi_Y(Y))$ is computed under the law $Y \sim \int_{\mathcal{X}} P_{\text{chan}}(\cdot \mid x) d\Pi_X(x)$ and the conditional entropy is over the joint law of $(\varphi_X(X), \varphi_Y(Y))$ induced by Π_X and P_{chan} .

The definition is descriptor-relative: it measures information transfer between the *declared descriptors* φ_X and φ_Y , not between raw X and $\varphi_Y(Y)$. Conditioning on full X would leak structure outside the declared descriptor and undermine the framework’s descriptor-relativity claims (§3); the form above is consistent with the empirical estimator, which takes $(\varphi_X(X), \varphi_Y(Y))$ -paired samples and never reads X outside φ_X .

Proposition 2 (Population MI bound). $I_{\text{struct}}^{\text{pop}}(\mathcal{E}) \leq H(\varphi_X(X)) \leq H(\Pi_X)$.

Proof. The first inequality is $I(\varphi_X(X); \varphi_Y(Y)) \leq H(\varphi_X(X))$, the standard MI-entropy bound. The second is data-processing applied to φ_X as a deterministic post-processing of X : $H(\varphi_X(X)) \leq H(X) = H(\Pi_X)$. $\square$

Corollary 3 (Fixed-template degeneracy). When $\mathcal{X} = \{x_0\}$ is a single fixed template, $H(\Pi_X) = 0$, so $I_{\text{struct}}^{\text{pop}}(\mathcal{E}) = 0$ by construction.

This corollary is the formal content of the framework’s claim that biological systems with single fixed templates (Drt3a’s single ncRNA, a single chromosome being replicated within a cell) have zero population MI. Population MI measures *cross-realization information transfer*; for a fixed template, there are no realizations to transfer between.

1.3 Substrate capacity

The capacity of a templating substrate is defined as a supremum of population MI over input ensembles. Because the framework’s heredity claim concerns *recursive* information transfer (the product must itself be readable as a next-round template), the supremum is taken over ensembles supported on the recursively closed set:

$$\mathcal{R}_L := \{X \in \mathcal{X}_L : \mathcal{O}^t(X) \in \mathcal{X}_L \text{ for all } t \geq 0, \text{ possibly via the reversible involution } \iota\}.$$

Restricting the supremum to $\mathcal{R}_L$ makes R3 (catalytic closure under iteration) mandatory in the capacity definition rather than argued for separately in the proof. Substrates failing R3 (e.g., translation, where the protein product is not ribosome-readable) have $\mathcal{R}_L = \emptyset$ or trivially small, so their substrate capacity is zero or bounded by a single-application transfer.

Definition 4 (Substrate capacity, recursive). For a templating substrate class $\mathcal{S}$ and template length L , the substrate capacity is

$$C_{\mathcal{S}}(L) := \sup_{\Pi_X \text{ on } \mathcal{R}_L} I_{\text{struct}}^{\text{pop}}(\mathcal{E}).$$

The supremum is over template-distribution choices on the recursively closed input set $\mathcal{R}_L \subseteq \mathcal{X}_L$. $C_{\mathcal{S}}$ is a property of the substrate class, not of any particular biological realization. In contrast, $I_{\text{struct}}^{\text{pop}}$ is the realized population MI in a specified system. Both are needed; conflating them is the most common failure mode of substrate-level reasoning.

Remark 5 (On the choice to restrict to $\mathcal{R}_L$). A reader might prefer to define a separate one-step channel capacity over all of $\mathcal{X}_L$. That object is well-defined and useful for non-recursive transfers (e.g., translation), but it cannot ground a *recursive* heredity claim: a substrate with $\Omega(L)$ one-step capacity but $\mathcal{R}_L = \emptyset$ transmits information one time and then exits the substrate class. Heredity, in the framework’s sense, requires the iteration to remain within $(\mathcal{X}, \mathcal{O})$, and this is exactly what restricting to $\mathcal{R}_L$ encodes.

Remark 6. For Mode 1 (Watson-Crick) at alphabet size $|\mathcal{A}|$ and per-position misincorporation rate ε , $C_{\mathcal{S}}(L) = L \cdot [\log_2 |\mathcal{A}| - h_b(\varepsilon) - \varepsilon \log_2(|\mathcal{A}| - 1)]$, achieved by the uniform distribution Π_X^{unif} on $\mathcal{R}_L$ (which for Mode 1 equals $\mathcal{X}_L$ up to involution-equivalence). For Mode 3 with N -state cyclic dynamics, $C_{\mathcal{S}}(L) \leq \log_2 N$ for all L , regardless of template-distribution choice (Theorem 14).

1.4 Channel mutual information

Definition 7 (Channel MI against an ensemble). Let $\mathcal{C} = \{c_1, \dots, c_K\}$ be a finite ensemble of templating channels with prior $P(C)$. For a focal channel $c \in \mathcal{C}$,

$$I_{\text{struct}}^{\text{chan}}(c; \mathcal{C}) := D_{\text{KL}}(P(\varphi_Y(Y) \mid C = c) \parallel \bar{P}(\varphi_Y(Y)))$$

where the mixture distribution is

$$\bar{P}(\varphi_Y(Y)) := \sum_{c' \in \mathcal{C}} P(C = c') P(\varphi_Y(Y) \mid C = c').$$

The full ensemble channel MI is

$$\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) := \sum_{c \in \mathcal{C}} P(C = c) D_{\text{KL}}(P(\varphi_Y(Y) \mid C = c) \parallel \bar{P}).$$

Proposition 8 (Channel MI bound). $\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) \leq \log_2 |\mathcal{C}|$, with equality iff the channels in $\mathcal{C}$ produce mutually disjoint output distributions.

Proof. $\bar{I}_{\text{struct}}^{\text{chan}}(\mathcal{C}) = I(C; Y)$ under the joint law $P(C, Y) = P(C)P(Y \mid C)$. $H(C) \leq \log_2 |\mathcal{C}|$. The mutual information is bounded by either marginal entropy. Equality of $I(C; Y)$ and $H(C)$ occurs iff Y determines C , i.e., iff the conditional distributions $P(Y \mid C = c)$ have disjoint supports. $\square$

Remark 9 (Channel MI for fixed templates). Unlike $I_{\text{struct}}^{\text{pop}}$, $I_{\text{struct}}^{\text{chan}}$ does not vanish for fixed-template channels. A channel with a single fixed template still produces a distinguishable output distribution against alternative channels, and the divergence $D_{\text{KL}}(P(Y \mid C = c) \parallel \bar{P})$ measures that distinguishability. This is the formal content of the framework’s claim that $I_{\text{struct}}^{\text{chan}}$ distinguishes Drt3a (Mode 1 with fixed ncRNA template) from Drt3b (Mode 3 with cyclic active site) even when both are individual cellular realizations with fixed mechanisms.

1.5 Per-base mechanism fidelity

Definition 10 (Phase fidelity and monomer fidelity). For a templating channel with cyclic dynamics having N states, define a phase-trajectory function $\sigma : \mathbb{N} \rightarrow \{0, 1, \dots, N-1\}$ that gives the canonical phase for each position. The *phase fidelity* is

$$f_{\text{phase}}(c) := \mathbb{E} \left[\frac{1}{L} \sum_{i=1}^L \mathbf{1}\{\hat{\sigma}_i(Y) = \sigma(i)\} \right]$$

where $\hat{\sigma}_i$ is the phase recovered from position i of Y . The *monomer fidelity* at state s is

$$f_{\text{monomer}}(c; s) := \mathbb{E} \left[\frac{1}{|\{i : \sigma(i) = s\}|} \sum_{i: \sigma(i)=s} \mathbf{1}\{Y_i = \pi(s)\} \right]$$

where $\pi(s)$ is the canonical monomer for state s . For acyclic channels (Mode 1, Mode 5), $N = 1$, $f_{\text{phase}} \equiv 1$, and only f_{monomer} is meaningful, simplifying to the per-position correctness rate.

Remark 11. f_{phase} and f_{monomer} are not mutual informations. They are moments of the joint distribution capturing different mechanism-level observables. The split is necessary because cyclic channels can fail in two structurally distinct ways: an architectural disruption that breaks f_{phase} (the cycle no longer alternates) versus a selectivity disruption that degrades f_{monomer} within an intact cycle (the alternation continues but with degraded monomer choice). These correspond to the architectural-versus-selectivity gate distinction reported in main paper Results.

1.6 The apparatus signature

Definition 12 (Joint signature). The apparatus signature for a templating channel c against an ensemble $\mathcal{C}$ is the tuple

$$\sigma(c; \mathcal{C}) := \left(I_{\text{struct}}^{\text{pop}}(c), I_{\text{struct}}^{\text{chan}}(c; \mathcal{C}), f_{\text{phase}}(c), f_{\text{monomer}}(c; s) \right)_{s=0}^{N-1}.$$

Mode classification uses the joint signature; signature collapse to any single coordinate is information-discarding.

2 Mode-specific information bounds

This section proves the scaling-law claims for each of the five primary modes. Mode 6 is treated separately in §2.6.

2.1 Mode 1 — linear scaling

Theorem 13 (Mode 1 capacity). *For a Watson-Crick channel with alphabet $|\mathcal{A}| \geq 2$, per-position misincorporation rate ε , template population $\mathcal{X} = \mathcal{A}^L$ with uniform distribution Π_X^{unif} , and product Y of length L ,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) = L \cdot [\log_2 |\mathcal{A}| - h_b(\varepsilon) - \varepsilon \log_2 (|\mathcal{A}| - 1)].$$

Proof. The Watson-Crick channel is discrete memoryless: $P(Y_i = \text{wc}(x_i) \mid X_i = x_i) = 1 - \varepsilon$, and $P(Y_i = a' \mid X_i = x_i) = \varepsilon / (|\mathcal{A}| - 1)$ for each $a' \neq \text{wc}(x_i)$. By memorylessness, the channel decomposes per position. For any position i , $I(X_i; Y_i) = H(X_i) - H(X_i \mid Y_i)$. Under uniform Π_X , $H(X_i) = \log_2 |\mathcal{A}|$. By Bayes' rule and uniformity, $H(X_i \mid Y_i) = H(Y_i \mid X_i) = h_b(\varepsilon) + \varepsilon \log_2 (|\mathcal{A}| - 1)$. Summing over L positions gives the stated formula. $\square$

Empirical match. Across 30 sweep cells with $\varepsilon \in \{0.001, 0.01, 0.05, 0.10, 0.25\}$ and $L \in \{10, 25, 50, 100, 200, 500\}$, $n_{\text{samples}} = 5000$ per cell, the empirical $I_{\text{struct}}^{\text{pop}}$ matches Theorem 13 within 0.6% across all cells (Test A.1, main paper Fig. 1A).

2.2 Mode 3 — logarithmic saturation

Theorem 14 (Mode 3 capacity bound). *For an N -state cyclic templating channel with phase $X \in \{0, 1, \dots, N - 1\}$ drawn uniformly, and product Y of length L generated by deterministic cycle progression with per-step substrate-selection error ε ,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq \log_2 N$$

for all L , provided the N phase states are pairwise distinguishable under the product descriptor φ_Y (so that the phase-conditioned output distributions $P(\varphi_Y(Y) \mid X = j)$ are pairwise distinct). The bound is approached in the low-noise long-block limit ($L \rightarrow \infty$ with ε fixed at small positive value); the bound is saturated exactly at $\varepsilon = 0$.

Proof. By the data-processing inequality applied to the cycle progression (which is a deterministic function of phase and time step), $I(X; Y) \leq I(X; \text{phase trajectory}) \leq H(X) = \log_2 N$. The bound is achieved iff phase is recoverable from Y , which requires the phase-conditioned output distributions to be pairwise distinguishable. If two phase states $j \neq j'$ produce identical output distributions under φ_Y , then $I(X; Y) < \log_2 N$ strictly. Under pairwise distinguishability and $\varepsilon = 0$, phase is recoverable from Y_1 alone (the first position determines the phase that produced it). At $\varepsilon > 0$ the recoverability is imperfect on a single position but improves with L because the periodic structure of Y provides redundant phase information. Standard channel-coding asymptotic analysis (Cover and Thomas, 2006, Theorem 8.7.1) gives $\lim_{L \rightarrow \infty} I(X; Y) = \log_2 N$ for fixed $\varepsilon > 0$ in the regime where the cycle's per-step Shannon capacity exceeds zero. $\square$

Empirical match. For $N \in \{2, 3, 4, 5, 6, 8, 10\}$, $\varepsilon = 10^{-3}$, L from $L = N$ to $L = 50N$: empirical $I_{\text{struct}}^{\text{pop}}$ approaches $\log_2 N$ to within estimator precision (drift across L values $< 0.0001\%$; Test B, main paper Fig. 2A,B).

2.3 Mode 5 — module-bounded

Theorem 15 (Mode 5 capacity). *For a modular conveyor with N modules, each independently selecting one monomer from substrate alphabet of size k with fidelity $1 - \varepsilon$, and product Y of length L :*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq \min(L, N) \cdot [\log_2 k - h_b(\varepsilon) - \varepsilon \log_2(k - 1)],$$

with positions beyond $\min(L, N)$ contributing zero information (templated positions saturate at N ; untemplated positions in a $L > N$ product carry no per-position information about the modular template).

Proof. Each module independently determines one product position; modules 1 through $\min(L, N)$ are templated, positions $N + 1, \dots, L$ (when $L > N$) have no module association and are drawn from a fixed distribution independent of the modular template. Per-position MI follows the same memoryless argument as Theorem 13 for templated positions and equals 0 for non-templated positions. Summing over the $\min(L, N)$ templated positions gives the stated bound. $\square$

Empirical match. Test C, main paper Fig. 2D.

2.4 Mode 2 — code-mediated capacity ceiling

Theorem 16 (Mode 2 capacity bound). *For code-mediated translation with DNA template of L codons drawn uniformly random from the 64 codons, standard genetic code (NCBI table 1), translation error rate $\varepsilon \ll 1$, and peptide output Y of length L over 20 amino acids plus stop,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq L \cdot H(\nu),$$

where ν is the codon-count distribution: $\nu(a) = n(a)/64$, with $n(a)$ the number of codons mapping to amino acid a in the standard genetic code (e.g., $n(\text{Leu}) = 6$, $n(\text{Met}) = 1$).

Proof. $I(X; Y) \leq H(Y)$. For uniform random X , the marginal distribution of Y_i (per codon) is exactly ν , so $H(Y_i) = H(\nu)$. By memorylessness of per-codon translation, $H(Y) = L \cdot H(\nu)$. $\square$

Empirical match. $H(\nu) = 4.218$ bits/codon for the standard genetic code (computed from the codon table, including stop as a 21st symbol); empirical $I_{\text{struct}}^{\text{pop}}$ at $L = 64$, $n_{\text{samples}} = 1000$, gives 4.214 bits/codon (Test F4). Match to four significant figures.

Capacity comparison with Mode 1. The same DNA at length $3L$ nucleotides has Mode 1 capacity $6L$ bits (Theorem 13 at $|\mathcal{A}| = 4$). Mode 2 capacity is $L \cdot H(\nu) = 4.218L$ bits. Capacity ratio $= 6/4.218 \approx 1.42$. Code degeneracy reduces information transfer by approximately 30% from the substrate ceiling.

Note on alternative bounds. The frequently-cited $L \cdot \log_2(20) = 4.32L$ bits assumes uniform marginal over amino acids, which the standard genetic code does not produce; the correct bound uses $H(\nu)$. Including stop as a 21st symbol gives $H(\nu) = 4.218$ bits; restricting to the 61 sense codons only gives $H(\nu_{\text{sense}}) \approx 4.139$ bits.

Low-error caveat. The bound $L \cdot H(\nu)$ is the no-error / low-error ceiling: it assumes translation fidelity is high enough that codon-to-amino-acid assignment is effectively deterministic. With non-negligible mistranslation rate $\varepsilon_t > 0$ per codon, the per-codon channel introduces a noisy mapping with capacity strictly below $H(\nu)$, and the realized $I_{\text{struct}}^{\text{pop}}$ is correspondingly lower. The theorem above states the ceiling; including mistranslation requires a per-codon channel-capacity correction analogous to the $h_b(\varepsilon)$ term in Theorem 13.

2.5 Mode 4 — length independence

Theorem 17 (Mode 4 capacity bound). *For a conformer-state templating channel with K distinguishable conformers and templated polymers of length L ,*

$$I_{\text{struct}}^{\text{pop}}(X; Y) \leq \log_2 K$$

independently of L .

Proof. X (the templating conformer) takes K values; $H(X) \leq \log_2 K$. The product Y inherits the conformer assignment as a single discrete attribute (its conformational state), regardless of polymer length. The polymer’s primary structure is fixed by the genome, not by the conformer. $\square$

Remark 18. Mode 4 is “state-templating” rather than “position-templating”; the length-independence is a property of the substrate (conformational identity), not a degenerate special case of position templating. The framework treats Mode 4 as $O(1)$ information content as a structural feature, not as an asymptotic limit.

2.6 Mode 6 — surface-position capacity is possible; biological instances are parasitic and R4-limited

For a 2D surface-position templating system with N distinguishable surface positions and per-position selection from k options, the combinatorial capacity bound is

$$C_S^{\text{combinatorial}}(N) \leq N \cdot \log_2 k,$$

which can in principle scale unboundedly as the surface area grows. Mode 6 capacity scaling is therefore not bounded by dimensionality alone.

The framework’s distinguishing claim about Mode 6 is that biological instances fail two operational requirements distinct from raw combinatorial capacity:

1. **Parasitic copyability.** The operator that copies the surface pattern, and the units that the surface assembles, are both produced by Mode 1 acting on the genome. Failure of Mode 1 inheritance terminates Mode 6 inheritance. Mode 6’s apparent capacity is gated by Mode 1’s substrate.
2. **Bounded heritable variation.** The variation that biological Mode 6 transmits is geometric perturbation of an existing unit repertoire, not compositional novelty. The number of distinguishable cortical states a lineage can transmit is bounded by combinatorial arrangements of genome-encoded parts, not by the surface’s combinatorial position count. Open-ended scaling requires unboundedly many heritable distinguishable states; biological Mode 6 lacks this because new compositional novelty would require new genome-encoded parts.

We omit a formal bound because Mode 6’s biological instances (ciliate cortex) are not fully characterized at the apparatus level in the published literature. The operational implementation of bounded heredity in the H3 mechanism characterization is the empirically relevant claim; see §5 below.

3 Descriptor-relativity

Definition 19. Two descriptor maps φ_Y, φ'_Y on $\mathcal{Y}$ are *equivalent* if there exists a bijection π such that $\varphi'_Y = \pi \circ \varphi_Y$. A templating event $\mathcal{E}$ analyzed under non-equivalent descriptors φ_Y and φ'_Y may give different $I_{\text{struct}}^{\text{pop}}$ values.

Theorem 20 (Descriptor non-uniqueness for designed-replicator templating). *Let a templating event $\mathcal{E}$ propagate constraints between a parent and daughter unit through residue-pairing chemistry (e.g., a designed β -sheet peptide replicator in which residues at paired positions select their partners by side-chain complementarity). Under a descriptor φ_Y^{pair} that records residue identity at each paired position, the templating event classifies as Mode-1-like with information bound determined by the pairing constraints, and $I_{\text{struct}}^{\text{pop}}$ scales with the number of paired positions. Under a descriptor φ_Y^{conf} that records only the conformational state (which β -sheet topology was adopted), the same templating event has $I_{\text{struct}}^{\text{pop}} = O(1)$.*

Sketch. Under φ_Y^{pair} , the joint distribution of daughter-unit residue identities given parent residue identities at paired positions has the structure of a position-by-position channel with per-position capacity determined by the pairing-rule alphabet; $I_{\text{struct}}^{\text{pop}}$ scales linearly with the number of paired positions, subject to those constraints. Under φ_Y^{conf} , only the conformational state is observed; the daughter inherits the parent’s conformation regardless of how many paired positions exist, so $I_{\text{struct}}^{\text{pop}}$ is bounded by $\log_2(\text{number of distinguishable conformations}) = O(1)$. $\square$

Remark 21 (Why not amyloid sequence-templating). A naive version of this point invokes amyloid or prion sequence propagation as the classifier-dependent case. That framing is incorrect: in biological prions, the amino-acid sequence is genome-encoded, and the amyloid does not template primary sequence at all. The phenomenon biological amyloids transmit is conformational — not sequence — and the framework treats this as a clean Mode 4 case (see Discussion, yeast-prion paragraph). The descriptor-relativity point made here is a property of templating systems where pairing-chemistry constraints *do* act position-by-position on the daughter unit’s primary structure; designed β -sheet replicators and analogous synthetic systems are the cleanest examples.

Corollary 22. *Mode classification applies to (templating event, descriptor) pairs, not to physical objects. The same template can mediate templating events that classify into different modes under different descriptors.*

This is what allows the framework to avoid an unresolvable taxonomic question of the form “is amyloid templating Mode 1 or Mode 4?” The question is ill-posed; the answer depends on what is being measured.

4 Bulk-matched controls as null hypotheses

Definition 23 (k -th order matched null). Given a product Y from a templating event, the k -th order matched null distribution Q_k is the maximum-entropy distribution on $\mathcal{Y}$ that preserves all k -th order marginal statistics of the empirical distribution of Y . Specifically: Q_1 preserves single-position marginals (composition) and is iid from those marginals; Q_2 preserves nearest-neighbor frequencies and is the maximum-entropy first-order Markov chain consistent with those frequencies; Q_k preserves k -position joint marginals and is the maximum-entropy $(k - 1)$ -th order Markov chain.

Remark 24. For $k = 2$, Q_2 is *not* iid; samples from Q_2 are first-order Markov sequences whose nearest-neighbor frequencies match the empirical Y .

Definition 25 (Bulk-matched paired null). The bulk-matched paired null for a templating event $\mathcal{E}$ is constructed by holding the template ensemble fixed and replacing the channel output with a draw from a k -th order matched distribution that destroys position-specific template-product correspondence. Specifically, for each observed template X , the null product Y_{bulk} is drawn from

$$Y_{\text{bulk}} \sim Q_k(\cdot \mid \text{compositional class of } X)$$

where Q_k is the k -th order matched distribution fitted to the product ensemble, optionally conditioned on the low-order composition class associated with X but *not* on X 's position-specific structure. This preserves k -th order marginal statistics between X and Y_{bulk} while destroying the templating correspondence.

Definition 26 (Structure-scrambled null). The structure-scrambled null is a second-order companion to Definition 25: for each (X, Y) pair, replace Y with Y_{scr} , a positionally permuted copy of Y that preserves the compositional histogram but randomizes the position assignment. Position-specific information transfer from X to Y is destroyed; compositional information is preserved.

Proposition 27 (Bulk and structure-scrambled controls as nulls). *A claim that $\mathcal{E}$ is a templating event with position-specific structural information transfer requires*

$$I_{\text{struct}}^{\text{pop}}(\varphi_X(X); \varphi_Y(Y)) > \tau \cdot I_{\text{struct}}^{\text{pop}}(\varphi_X(X); \varphi_Y(Y_{\text{bulk}}))$$

and

$$I_{\text{struct}}^{\text{pop}}(\varphi_X(X); \varphi_Y(Y)) > \tau \cdot I_{\text{struct}}^{\text{pop}}(\varphi_X(X); \varphi_Y(Y_{\text{scr}}))$$

where τ is a stated significance threshold (the main paper uses $\tau \geq 20$ for a “pass” verdict).

The threshold τ encodes the strength of the claim. The paired-null formulation is essential: in an earlier formulation, X_{bulk} and Y_{bulk} were drawn independently, which makes $I = 0$ exactly (modulo finite-sample estimator bias), giving a useless control. The paired null keeps the template marginal but breaks the position-specific correspondence, so any residual mutual information reflects compositional or low-order structure shared between X and the product ensemble. Whether a k -th order match is sufficient depends on what k captures the confounders: for Mode 1 versus random output, $k = 1$ is sufficient; for Mode 3 versus Mode 1 with periodic-like template, $k \geq 2$ is needed because nearest-neighbor structure differs. The framework reports controls with $k = 2$ by default.

5 Capacity and generation theorems

This section formalizes the two necessity results referenced in the main paper: the *capacity theorem* (R1, R2, R3 are necessary for unlimited heredity capacity) and the *generation theorem* (R4 is additionally necessary for cumulative generative recursion beyond deterministic closure). The structure is: notation and structural conditions (§5.1); formal definition of heredity capacity with its upper bound (§5.2); formal definition of deterministic closure and cumulative generative recursion (§5.3); the capacity theorem with proof (§5.4); the generation theorem with proof (§5.5); the bounded-heredity corollary recovering each mode’s capacity ceiling (§5.6); the finite-population, finite-horizon proposition (§5.7); and the recovery of prior formulations within this structure (§5.8).

5.1 Notation and structural conditions

Let $\mathcal{X}_L$ denote the space of admissible configurations at size parameter L ; this is countable in all biological examples we treat. Let Σ be a finite alphabet, called the *recognition alphabet* of the catalytic operation. A descriptor is a map $\varphi_X : \mathcal{X}_L \rightarrow \Sigma^{N(L)}$ for some position count $N(L) \geq 1$. The catalytic operation $\mathcal{O}$ has kernel $Q_L : \mathcal{X}_L \times \mathcal{X}_L \rightarrow [0, 1]$ giving the conditional probability of producing $X' = \mathcal{O}(X)$ from template X :

$$Q_L(X' | X) \geq 0, \quad \sum_{X' \in \mathcal{X}_L} Q_L(X' | X) = 1 \text{ for all } X. \quad (1)$$

We allow $\mathcal{O}$ to be composed with a reversible involution $\iota : \mathcal{X}_L \rightarrow \mathcal{X}_L$ (e.g., Watson–Crick complementation), with the recursion taking the form $X \rightarrow \iota(X') \rightarrow X'' \rightarrow \dots$; for notational simplicity we suppress ι where it does not affect the argument.

The five conditions of the substrate recursion framework are:

(R1) Recognition-alphabet entropy. $|\Sigma| \geq 2$, so $\log_2 |\Sigma| > 0$.

(R2) Unbounded extensible recognition. $\liminf_{L \rightarrow \infty} N(L) = \infty$, where $N(L)$ is the count of independently variable recognized positions of the descriptor φ_X under Q_L . “Independently variable” is meant in the first-order, descriptor-relative sense: the substrate is not constrained to a finite recurrent phase repertoire, and increasing L increases the count of heritable recognized degrees of freedom. The condition does not require strong joint-statistical independence across positions; context-dependent error rates (as in real DNA polymerases) are permitted.

(R3) Catalytic closure under iteration. For every $X \in \mathcal{X}_L$ in the support of Q_L , $\mathcal{O}(X) \in \mathcal{X}_L$ (possibly via the reversible involution ι), so that $\mathcal{O}^2(X)$ is well-defined and the iteration $X \rightarrow \mathcal{O}(X) \rightarrow \mathcal{O}^2(X) \rightarrow \dots$ remains within $\mathcal{X}_L$.

(R4) Stochastic drift in Σ . The kernel $Q_L(\cdot | X)$ has $|\text{supp } Q_L(\cdot | X)| \geq 2$ for at least some $X \in \mathcal{X}_L$, with the dispersion of $\varphi_X(X')$ around $\varphi_X(X)$ lying in the recognition alphabet Σ of (R1). Drift in a separate alphabet $\Sigma' \neq \Sigma$ (e.g., chemical modifications outside the polymerase’s recognition palette) fails R4.

(R5) Scope: intrinsic catalysis. The kernel Q_L is determined by intrinsic physicochemistry — the active-site free-energy landscape, the available reactants, the chemical driving — not by an externally specified loss function or target product. R5 is scope-defining rather than structural: it demarcates the domain in which R1–R4 apply rather than introducing a fifth necessity condition.

5.2 Heredity capacity

Let Π_X denote any probability distribution on $\mathcal{X}_L$ (an input ensemble of templates), and let $X' \sim Q_L(\cdot | X)$ for $X \sim \Pi_X$.

Definition 28 (Heredity capacity). The *heredity capacity* of $(\mathcal{X}, \mathcal{O})$ at size L is

$$C_S(L) = \sup_{\Pi_X} I(\varphi_X(X); \varphi_X(X')), \quad (2)$$

the supremum of mutual information between input and output descriptors over all input ensembles.

Proposition 29 (Capacity upper bound). *For any kernel Q_L ,*

$$C_S(L) \leq N(L) \cdot \log_2 |\Sigma|. \quad (3)$$

Proof. By the standard identity $I(A; B) = H(B) - H(B | A) \leq H(B)$, for any input ensemble Π_X ,

$$I(\varphi_X(X); \varphi_X(X')) \leq H(\varphi_X(X')).$$

The descriptor $\varphi_X(X')$ takes values in $\Sigma^{N(L)}$, a set of cardinality $|\Sigma|^{N(L)}$, so

$$H(\varphi_X(X')) \leq \log_2 |\Sigma|^{N(L)} = N(L) \log_2 |\Sigma|.$$

The supremum over Π_X inherits the bound. $\square$

The bound is achievable iff some Π_X realizes a uniform $\varphi_X(X')$ on $\Sigma^{N(L)}$. Realizability requires that the recognition step actually resolve all $|\Sigma|$ states at each of $N(L)$ positions and that the iteration close within $\mathcal{X}_L$ — i.e., R1, R2, R3.

Definition 30 (Unlimited heredity capacity). $(\mathcal{X}, \mathcal{O})$ has *unlimited heredity capacity* if $C_S(L) = \Omega(L)$ as $L \rightarrow \infty$.

Remark 31 (Information-theoretic and combinatorial formulations). Definition 30 is the information-theoretic statement. Its combinatorial equivalent: in the discrete noiseless limit where the upper bound of Proposition 29 is saturated, the recursively readable configuration set $\mathcal{R}_L = \{X \in \mathcal{X}_L : \mathcal{O}^t(X) \in \mathcal{X}_L \text{ for all } t \geq 0\}$ satisfies $|\mathcal{R}_L| \geq 2^{cL}$ for some $c > 0$. The two formulations are coordinated through entropy: linear scaling of $C_S(L)$ in bits is equivalent to exponential scaling of the recursively readable state count. They describe the same fact at two levels.

5.3 Zero-drift closure and cumulative generation

A correct formal account of “what does R4 add beyond R1–R3” requires distinguishing the reachable set produced by the kernel’s stochastic dynamics from the reachable set produced by a single canonical deterministic skeleton. We previously formulated this distinction using a union over all deterministic restrictions of Q_L ; that formulation fails because, when Q_L is stochastic, the union absorbs every positive-probability branch and the resulting “closure” equals the full positive-probability reachable set, making strict containment logically impossible. The corrected formulation uses the canonical zero-drift map g_L , which is already implicit in the framework’s notion of a reference product X_{ref} .

Definition 32 (Zero-drift reference map). The *zero-drift reference map* of the catalytic operation $\mathcal{O}$ is the deterministic function $g_L : \mathcal{X}_L \rightarrow \mathcal{X}_L$ defined by

$$g_L(X) := X_{\text{ref}}(X),$$

where $X_{\text{ref}}(X)$ is the canonical product of $\mathcal{O}$ applied to template X under zero drift (the most-probable successor in the noise-free or low-noise limit; e.g., the Watson–Crick complement for a DNA polymerase, the exact alternating product for Drt3b). For all biological substrates we treat, g_L is well-defined; ties (if any) are broken canonically by the chemistry’s reference output.

Definition 33 (Zero-drift closure). The *zero-drift closure* of a set $S_0 \subset \mathcal{X}_L$ under $\mathcal{O}$ is

$$\text{cl}_0(S_0) = \bigcup_{t \geq 0} g_L^t(S_0). \quad (4)$$

Equivalently, $\text{cl}_0(S_0)$ is the orbit closure of S_0 under the deterministic zero-drift dynamics: the set of configurations reachable from S_0 if $\mathcal{O}$ produced its canonical reference output at every step with no error.

Definition 34 (Cumulative generative recursion). $(\mathcal{X}, \mathcal{O})$ supports *cumulative generative recursion* if, for every L sufficiently large and for every finite proper subset $S_0 \subsetneq \mathcal{X}_L$ with $|S_0| \ll |\mathcal{R}_L|$, the recursively readable descendant set

$$\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \{X \in \mathcal{X}_L : \exists X_0 \in S_0, t \geq 0 \text{ with } Q_L^t(X | X_0) > 0 \text{ and } \mathcal{O}^s(X) \in \mathcal{X}_L \text{ for all } s \geq 0\} \quad (5)$$

strictly contains $\text{cl}_0(S_0)$, with the additional configurations differing from elements of $\text{cl}_0(S_0)$ in local positions of the descriptor φ_X (i.e., in the alphabet Σ).

The zero-drift formulation does the work intended for the deterministic-closure definition without absorbing all stochastic branches. If Q_L is itself deterministic and equal to g_L (no drift), then $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \text{cl}_0(S_0)$, not strict containment — correctly characterizing R4-failure. If Q_L is stochastic with $\text{supp } Q_L(\cdot | X) \supsetneq \{g_L(X)\}$ for some X (i.e., R4 holds), then the descendant set contains off-orbit configurations not in $\text{cl}_0(S_0)$, and strict containment holds.

Remark 35 (Why this formulation is correct where the prior was not). The earlier formulation, $\text{cl}_{\text{det}}(S_0) = \bigcup_{f \in \mathcal{D}(Q_L)} \bigcup_t f^t(S_0)$ where $\mathcal{D}(Q_L)$ ranges over all deterministic restrictions of Q_L , fails because for any $X' \in \text{supp } Q_L(\cdot | X)$ there exists $f \in \mathcal{D}(Q_L)$ with $f(X) = X'$. Hence $\bigcup_{f \in \mathcal{D}(Q_L)} \{f(X)\} = \text{supp } Q_L(\cdot | X)$, and iterating gives $\text{cl}_{\text{det}}(S_0) =$ the full positive-probability reachable set. Strict containment $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) \supsetneq \text{cl}_{\text{det}}(S_0)$ is then impossible because $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) \subseteq \text{cl}_{\text{det}}(S_0)$ by construction. The zero-drift formulation picks a single canonical deterministic skeleton (g_L) rather than a union over all possible skeletons, restoring the desired strict-containment characterization of R4.

The counter-chain counterexample is still correctly excluded. A deterministic counter chain $\mathcal{X}_L = \{0, 1\}^L$, $\mathcal{O}(X) = X + 1 \pmod{2^L}$, has $Q_L = g_L$ (deterministic kernel coincides with its own zero-drift map), so $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \text{cl}_0(S_0)$ for any S_0 . The counter chain leaves S_0 but does not exceed $\text{cl}_0(S_0)$; cumulative generation fails. The mistake the earlier “leaves S_0 ” formulation made — treating non-trivial deterministic dynamics as generative — is correctly avoided.

5.4 Capacity theorem

Theorem 36 (Capacity necessity). *Within the (R5) scope, if $(\mathcal{X}, \mathcal{O})$ has unlimited heredity capacity in the sense of Definition 30, then R1, R2, and R3 hold.*

Proof. We prove the contrapositive: failure of any of R1, R2, R3 forces $C_S(L)$ to scale strictly slower than $\Omega(L)$.

Case R1 fails. If $|\Sigma| = 1$, then $\log_2 |\Sigma| = 0$. By Proposition 29, $C_S(L) \leq N(L) \cdot 0 = 0$, so $C_S(L) \equiv 0$, not $\Omega(L)$.

Case R2 fails. If $\liminf_{L \rightarrow \infty} N(L) = N_{\max} < \infty$, then $N(L) \leq N_{\max}$ for all L sufficiently large. By Proposition 29, $C_S(L) \leq N_{\max} \log_2 |\Sigma|$, a constant in L . So $C_S(L) = O(1)$, not $\Omega(L)$.

Case R3 fails. If $\mathcal{O}(X) \notin \mathcal{X}_L$ for some X in the support of Q_L (and not recoverable via reversible involution ι), then for any input X encountered along the iteration, there exists $t \geq 1$ with $\mathcal{O}^t(X) \notin \mathcal{X}_L$, so $X \notin \mathcal{R}_L$. The recursively closed set $\mathcal{R}_L$ is therefore empty (or restricted to a trivially small set of cycles, depending on the specific failure mode). By Definition 4, $C_S(L) = \sup_{\Pi_X \text{ on } \mathcal{R}_L} I_{\text{struct}}^{\text{pop}}$ is taken over an empty or trivially small support set; the supremum is zero or bounded by a constant

in L . The substrate’s recursive heredity capacity, properly defined to exclude one-step transfers to non-recursive products, is therefore not $\Omega(L)$.

By contrapositive, $C_S(L) = \Omega(L)$ requires R1, R2, and R3. $\square$

Remark 37 (R3 and one-step capacity). Theorem 36 is correct because the capacity is taken over Π_X supported on $\mathcal{R}_L$, not all of $\mathcal{X}_L$. A reader familiar with channel-coding might object that one-step transfer can have $\Omega(L)$ mutual information even when R3 fails (translation: DNA $\rightarrow$ protein, $\Omega(L)$ bits in a single round). This is correct as a statement about one-step transfer but not about recursive heredity. The framework’s claim concerns the substrate’s capacity to host *iterated* information transfer within its own substrate class, which is exactly the restriction encoded in $\mathcal{R}_L$.

Remark 38 (R4 is not required for capacity). Theorem 36 does not invoke R4. A perfect copier — $Q_L(X' | X) = \delta_{X',X}$, no drift — trivially satisfies $C_S(L) = N(L) \log_2 |\Sigma|$: the channel is the identity on $\mathcal{X}_L$, so $I(\varphi_X(X); \varphi_X(X')) = H(\varphi_X(X))$, and choosing Π_X uniform on $\mathcal{X}_L$ achieves $H(\varphi_X(X)) = N(L) \log_2 |\Sigma|$. This is the structural reason a hypothetical drift-free DNA polymerase would still be a high-capacity inheritance carrier — it would not innovate from a finite seed, but it would store and transmit the full 4^L -fold configurational repertoire faithfully.

5.5 Generation theorem

Theorem 39 (Generation necessity). *Within the (R5) scope, if $(\mathcal{X}, \mathcal{O})$ supports cumulative generative recursion in the sense of Definition 34, then R4 holds, in addition to the R1, R2, R3 inherited from coherence of the catalytic process.*

Proof. R1, R2, R3 are preconditions for cumulative generation as defined: cumulative generation requires a recognition alphabet (R1), at least one extensible position (R2 in the recognition-relevant sense), and a recursion that closes within $\mathcal{X}_L$ (R3). We focus on R4.

Suppose R4 fails. There are two cases.

Case (a): the kernel is deterministic and coincides with g_L . If $|\text{supp } Q_L(\cdot | X)| = 1$ for all $X \in \mathcal{X}_L$, then the unique positive-probability successor of X is $X_{\text{ref}}(X) = g_L(X)$ (by the definition of X_{ref} as the canonical zero-drift product). So Q_L on its support equals g_L , and every positive-probability trajectory of Q_L from S_0 is contained in the zero-drift orbit closure:

$$\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \bigcup_{t \geq 0} g_L^t(S_0) = \text{cl}_0(S_0).$$

This is not strict containment. Cumulative generation in the sense of Definition 34 fails.

Case (b): the kernel drifts in an alphabet $\Sigma' \neq \Sigma$. Suppose $Q_L(X' | X) > 0$ for some $X' \neq g_L(X)$, but the difference $\varphi_X(X') \neq \varphi_X(g_L(X))$ at position i involves an alphabet character $\sigma' \in \Sigma' \setminus \Sigma$. Then $\varphi_X(X')$ is not in $\Sigma^{N(L)}$ under the recognition map φ_X : the drift is invisible to the next iteration’s recognition step. The next $\mathcal{O}$ -application reads $\varphi_X(X')$ as if it lay in $\Sigma^{N(L)}$, treating σ' as the closest Σ -character or as undefined; either way, the drift in Σ' does not propagate as recursively readable variation. Hence X' with Σ' -drift is excluded from $\mathcal{R}_{Q_L}^{\text{rec}}(S_0)$ (the recursive-readability requirement). The descendant set therefore does not strictly contain $\text{cl}_0(S_0)$ *through branching local alternatives in Σ* (Definition 34’s qualifier).

In both cases, the cumulative-generation requirement fails. By contradiction, R4 holds whenever $(\mathcal{X}, \mathcal{O})$ supports cumulative generation. $\square$

Remark 40 (Counter-chain counterexample under the corrected definition). A deterministic counter chain illustrates why “leaves S_0 ” would have been insufficient and shows that the zero-drift formulation correctly excludes the counterexample. Take $\mathcal{X}_L = \{0, 1\}^L$, $S_0 = \{0^L\}$, and

$\mathcal{O}(X) = X + 1 \bmod 2^L$ with $g_L = \mathcal{O}$ (the kernel is deterministic and its canonical reference output is itself). Then $\text{cl}_0(S_0) = \bigcup_{t \geq 0} g_L^t(S_0) = \{0, 1, \dots, 2^L - 1\}$ already includes every state reachable along the orbit. Since $Q_L = g_L$, the descendant set satisfies $\mathcal{R}_{Q_L}^{\text{rec}}(S_0) = \text{cl}_0(S_0)$. Not strict containment. The counter chain leaves S_0 but does not exceed $\text{cl}_0(S_0)$; cumulative generation correctly fails.

5.6 Bounded-heredity corollary

The capacity and generation theorems give the bounded-heredity classification of substrates failing a strict subset of R1–R4.

Corollary 41 (Bounded heredity, mode-keyed). *Within the (R5) scope, substrates failing a strict subset of R1–R4 exhibit heredity bounded according to which condition fails:*

1. *Failure of R1 at the per-site level ($|\Sigma| = 1$): $C_S(L) = 0$. No heritable distinguishability per site.*
2. *Failure of R1 at the macro level via a finite attractor repertoire of size K , with no extensible recognition substructure: $C_S(L) \leq \log_2 K$, length-independent. Mode 4: prion strains, amyloid attractors, DRT1 filament conformer.*
3. *Failure of R2 via a finite recurrent phase repertoire of N states: $C_S(L) \leq \log_2 N$, regardless of product length. Mode 3: Drt3b at $N = 2$ gives $C_S(L) \leq 1$ bit.*
4. *Failure of R2 via finite module count N with per-module alphabet of size k : $C_S(L) \leq N \log_2 k$. Mode 5: NRPS, polyketide synthases.*
5. *Failure of R3: substrate-level heredity is bounded by the information transmitted in a single application of $\mathcal{O}$; recursion is parasitic on a separate substrate–operator pair that itself satisfies R3. Mode 2 (translation: protein not ribosome-readable; recursion parasitic on Mode 1 acting on the gene encoding the ribosome). Mode 6 (cortex: surface does not autonomously self-copy; recursion parasitic on Mode 1 acting on the genome).*
6. *Failure of R4: substrate has heredity capacity but cannot generate novelty from a finite initial repertoire. Heredity is bounded by the founding-population repertoire.*

The corollary is a direct consequence of the upper bound in Proposition 29 together with the contrapositive structure of Theorems 36–39: each condition failure picks off a specific multiplicative or additive term in the bound. The biological assignment to Modes 1–6 follows the main paper’s taxonomy table.

5.7 Finite-population proposition

The bounded-heredity corollary is a substrate-level statement (the heredity capacity ceiling at the kernel level). The corresponding population-level statement — what happens when a substrate–operator pair with a given failure pattern hosts a finite population under selection — requires careful scoping. The universal-cap formulation (“mechanisms failing R3 or R4 are structurally capped at $F_k^* < F_{\max}$ under any population dynamics”) is too strong, because a wholesale-redraw mechanism with full support over $\mathcal{X}_L$ can sample any configuration by chance with infinite time. The proper finite-horizon, finite-population form follows.

Proposition 42 (Finite-horizon plateau under nonlocal or absent drift). *Let a population of size K evolve under selection sharpness β toward a target phenotype on a tested-class landscape (single-target Hamming distance, L -dimensional alphabet, static), over a horizon $T < \infty$. Then:*

1. *Mechanisms whose substrate-operator pair fails $R4$ at the individual level (exact-copy systems: M1 lineage-level fixation, M3 individual-level exact copy) cannot exceed the maximum fitness F_0^* present in the founding lineage repertoire. Mean fitness at horizon T is bounded by F_0^* regardless of K , β , or T .*
2. *Mechanisms whose substrate-operator pair has nonlocal $R4$ (wholesale redraw: M2 at rate $r > 0$) sample novelty with full support over $\mathcal{X}_L$ but do not accumulate local improvements. Under the tested update rules, the probability of hitting the target X^* within horizon T is bounded by*

$$P(\tau \leq T) \leq 1 - \exp\left[-\frac{rKT}{|\mathcal{X}_L|}\right],$$

with expected hitting time $\mathbb{E}[\tau] \sim |\mathcal{X}_L|/(rK)$. Since $|\mathcal{X}_L|$ scales as $|\mathcal{A}|^L$, the expected waiting time scales exponentially in L , and is negligible at biological scales for any fixed K , r , and biologically plausible T .

3. *Mechanisms whose substrate-operator pair satisfies $R3$ and $R4$ with individual-level local drift ($M4$) are not structurally barred from cumulative climbing toward $F_{\max}$. Whether climbing actually occurs depends on landscape topology, finite-population drift, mutation load, neutral-network connectivity, and other conditions outside the framework's scope.*

The proposition is empirically tested in the main paper's Test H3–H5 sweep. The tested class ($L_{\text{target}} \in \{16, 32, 64, 128\}$, $K = 400$, $\beta \in \{2, 5, 10, 20\}$, 30 replicates per cell) confirms each item: M0 (no recursion) plateaus at chance (0.250); M1 and M3 ($R4$ fails) plateau near $F_0^* \approx 0.47$ –0.50; M2 at sweet-spot $r^* = 0.10$ plateaus at ~ 0.53 ; M4 reaches ~ 0.97 . The proposition's positive direction (M4 not structurally barred) is bounded: the simulation shows climbing on the tested landscape; it does not establish climbing on arbitrary landscapes.

5.8 Synthesis and recovery of prior formulations

The capacity and generation theorems clarify how prior formulations of the inheritance question fit within the framework. The correspondence in the table below is *conceptual* rather than mathematical equivalence: some entries are tight (Szathmáry's unlimited heredity is exactly $C_S(L) = \Omega(L)$ once the recursive-capacity definition is in place; Hull's replicator ontology is exactly R3 in its operational content), while others are interpretive in the sense that they identify which of the framework's conditions the prior formulation is addressing without claiming the two are formally equivalent statements (this is the appropriate reading of Pross's dynamic kinetic stability and Walker–Davies's top-down causation).

Prior formulation	Conceptual correspondence	Reference
Schrödinger’s aperiodic crystal	R1 + R2 (extensible recognition)	(Schrödinger, 1944)
Szathmáry’s unlimited heredity	$C_S(L) = \Omega(L)$ (capacity theorem)	(Szathmáry, 2000)
Eigen’s error threshold	Dynamical bound on Q_L given R4, R5	(Eigen, 1971; Eigen et al., 1989)
Wagner’s neutral networks	Topological condition on R4 drift in Σ	(Wagner, 2005, 2011)
Pross’s dynamic kinetic stability	Thermodynamic-driving condition on R4	(Pross, 2011)
Hull’s replicator ontology	R3 (substrate reads its own product)	(Hull, 1980)
Maynard Smith’s units of selection	R3 + R4 jointly	(Maynard Smith, 2000)
Hofmeyr’s (F, A) -systems	R3 in categorical form	(Hofmeyr, 2017)
Vasas et al. GARD refutation	Capacity-R2 failure for composition substrates	(Vasas et al., 2010)
Walker–Davies top-down causation	R3 in causal-flow form	(Walker and Davies, 2013)

The substantive contribution beyond these prior formulations is twofold: (i) the separation of *capacity* (R1, R2, R3) from *generation* (R4), which makes precise why a perfect copier transmits but does not innovate, and (ii) the explicit scope-defining role of R5, which marks the boundary between intrinsic catalytic chemistry and externally optimized search procedures.

Within this synthesis: Schrödinger’s aperiodic-crystal intuition is precisely R1 (alphabet entropy) and R2 (extensible recognition) jointly; Szathmáry’s unlimited heredity is exactly the capacity-theorem conclusion $C_S(L) = \Omega(L)$; Eigen’s error threshold is a dynamical bound *on* Q_L that becomes well-defined only once R4 holds and (R5) places the kernel in the intrinsic-physicochemistry domain; Wagner’s neutral-network requirement is the topological structure of drift that R4 produces; Pross’s dynamic kinetic stability supplies the thermodynamic driving under which R4 is sustainable; Hull’s replicator ontology requires R3 (the substrate’s product must be itself a substrate); Hofmeyr’s (F, A) -systems give the categorical formulation of R3; Vasas et al.’s GARD refutation independently establishes capacity-theorem R2 failure for composition-vector substrates via a dynamical-fidelity route; Walker–Davies top-down causation supplies the causal-flow content of R3 in the information-substrate-to-phenotype direction.

The framework’s value lies in the unification: each prior formulation addresses a particular condition or pair of conditions; the capacity/generation split shows precisely how the conditions combine and which biological chemistries fail which.

References

- T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, NJ, 2nd edition, 2006. ISBN 978-0-471-24195-9.
- M. Eigen. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58:465–523, 1971. doi: 10.1007/BF00623322.
- M. Eigen, J. McCaskill, and P. Schuster. The molecular quasi-species. *Advances in Chemical Physics*, 75:149–263, 1989.
- J.-H. S. Hofmeyr. Basic biological anticipation. In R. Poli, editor, *Handbook of Anticipation*. Springer, 2017.
- D. L. Hull. Individuality and selection. *Annual Review of Ecology and Systematics*, 11:311–332, 1980. doi: 10.1146/annurev.es.11.110180.001523.

- J. Maynard Smith. The concept of information in biology. *Philosophy of Science*, 67:177–194, 2000. doi: 10.1086/392768.
- A. Pross. Toward a general theory of evolution: extending Darwinian theory to inanimate matter. *Journal of Systems Chemistry*, 2:1, 2011. doi: 10.1186/1759-2208-2-1.
- E. Schrödinger. *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.
- E. Szathmáry. The evolution of replicators. *Philosophical Transactions of the Royal Society B*, 355: 1669–1676, 2000. doi: 10.1098/rstb.2000.0730.
- V. Vasas, E. Szathmáry, and M. Santos. Lack of evolvability in self-sustaining autocatalytic networks constraints metabolism-first scenarios for the origin of life. *Proceedings of the National Academy of Sciences*, 107:1470–1475, 2010. doi: 10.1073/pnas.0912628107.
- A. Wagner. *Robustness and Evolvability in Living Systems*. Princeton University Press, Princeton, 2005. ISBN 978-0-691-12240-3.
- A. Wagner. *The Origins of Evolutionary Innovations: A Theory of Transformative Change in Living Systems*. Oxford University Press, Oxford, 2011. ISBN 978-0-19-969259-1.
- S. I. Walker and P. C. W. Davies. The algorithmic origins of life. *Journal of the Royal Society Interface*, 10(79):20120869, 2013. doi: 10.1098/rsif.2012.0869.